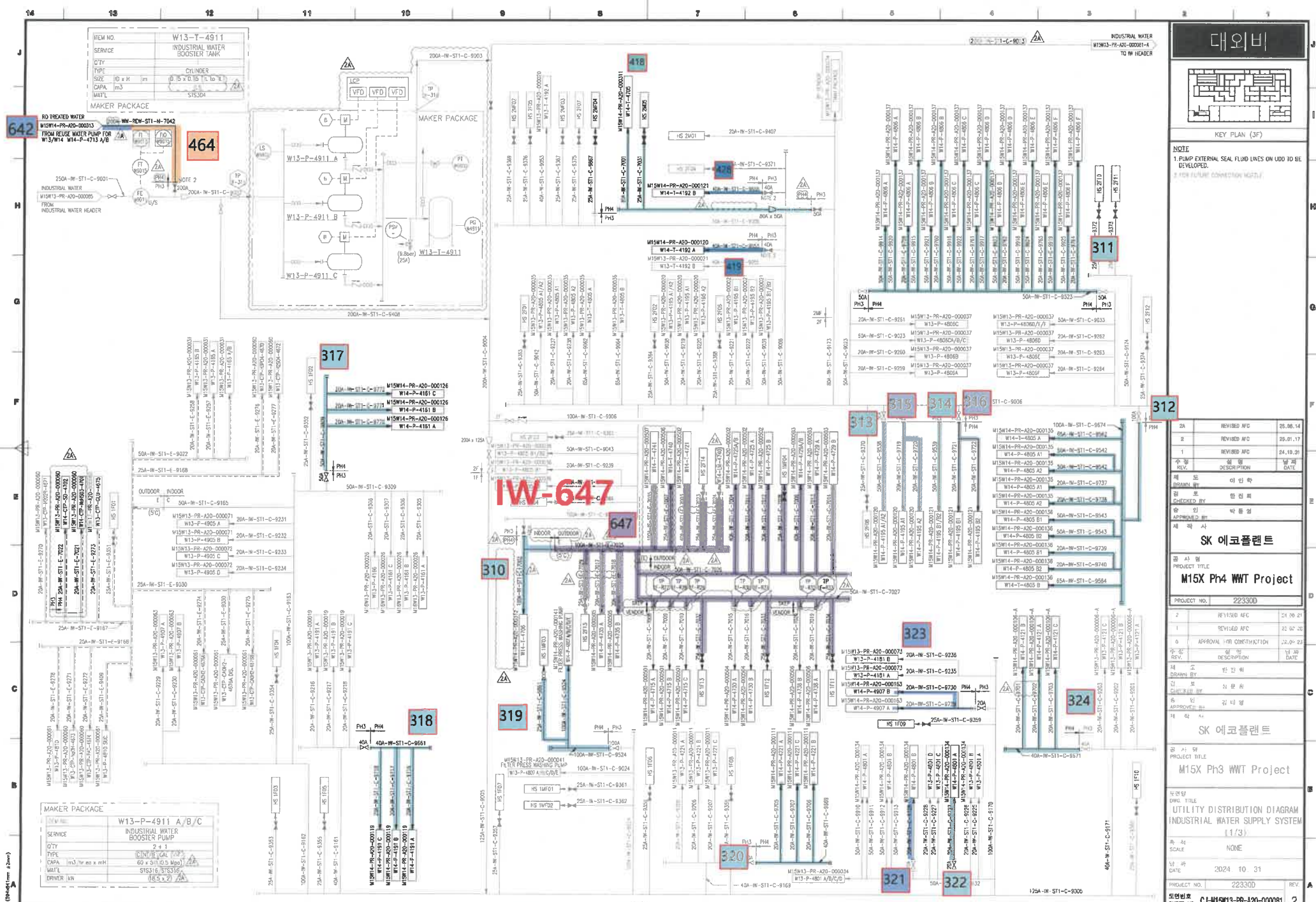


DRAWING LIST FOR TEST PACKAGE**IW-647****PROJECT NO.: 22328D****PROJECT TITLE : M15X Ph2 Project**

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	647	R1FA	20A-IW-ST1-C-7008-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
2	647	R1FA	20A-IW-ST1-C-7009-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
3	647	R1FA	20A-IW-ST1-C-7010-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
4	647	R1FA	20A-IW-ST1-C-7015-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
5	647	R1FA	20A-IW-ST1-C-7016-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
6	647	R1FA	20A-IW-ST1-C-7019-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
7	647	R1FA	20A-IW-ST1-C-7020-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
8	647	R1FA	25A-IW-ST1-C-7032-01	1 OF 1	0	CJ-M15W13-PR-A20-000081	2	
9	647	R1FA	25A-IW-ST1-C-7033-01	1 OF 1	0	CJ-M15W13-PR-A20-000081	2	
10	647	R1FA	25A-IW-ST1-C-7034-01	1 OF 1	0	CJ-M15W13-PR-A20-000081	2	
11	647	1FA	50A-IW-ST1-C-7027-01	1 OF 1	0	CJ-M15W13-PR-A20-000081	2	
12	647	R1MF	20A-IW-ST1-C-7011-01	1 OF 1	1	CJ-M15W14-PR-A20-000011	2A	
13	647	R1MF	20A-IW-ST1-C-7012-01	1 OF 1	0	CJ-M15W14-PR-A20-000011	2A	
14	647	R1MF	20A-IW-ST1-C-7013-01	1 OF 1	0	CJ-M15W14-PR-A20-000011	2A	
15	647	R1MF	20A-IW-ST1-C-7014-01	1 OF 1	0	CJ-M15W14-PR-A20-000011	2A	
16	647	R1MF	25A-IW-ST1-C-7028-01	1 OF 1	0	CJ-M15W13-PR-A20-000081	2	
17	647	R1MF	40A-IW-ST1-C-7004-01	1 OF 4	1	CJ-M15W13-PR-A20-000081	2	
18	647	R1MF	40A-IW-ST1-C-7004-02	2 OF 4	1	CJ-M15W13-PR-A20-000081	2	
19	647	R1MF	40A-IW-ST1-C-7004-03	3 OF 4	0	CJ-M15W13-PR-A20-000081	2	

DRAWING LIST FOR TEST PACKAGE						IW-647		
PROJECT NO.: 22328D								
PROJECT TITLE : M15X Ph2 Project								
No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
20	647	R1MF	40A-IW-ST1-C-7004-04	4 OF 4	0	CJ-M15W13-PR-A20-000081	2	
21	647	R1MF	40A-IW-ST1-C-7006-01	1 OF 4	1	CJ-M15W14-PR-A20-000502	1A	
22	647	R1MF	40A-IW-ST1-C-7006-02	2 OF 4	1	CJ-M15W14-PR-A20-000502	1A	
23	647	R1MF	40A-IW-ST1-C-7006-03	3 OF 4	0	CJ-M15W14-PR-A20-000502	1A	
24	647	R1MF	40A-IW-ST1-C-7006-04	4 OF 4	0	CJ-M15W14-PR-A20-000502	1A	
25	647	1MF	50A-IW-ST1-C-7026-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
26	647	1MF	50A-IW-ST1-C-7027-01	1 OF 1	0	CJ-M15W13-PR-A20-000081	2	
27	647	R2FA	20A-IW-ST1-C-7017-01	1 OF 1	1	CJ-M15W14-PR-A20-000011	2A	
28	647	R2FA	20A-IW-ST1-C-7018-01	1 OF 1	1	CJ-M15W14-PR-A20-000011	2A	
29	647	R2FA	25A-IW-ST1-E-7007-01	1 OF 1	0	CJ-M15W14-PR-A20-000506	1A	
30	647	R2FA	25A-IW-ST1-C-7023-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
31	647	R2FA	25A-IW-ST1-C-7024-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
32	647	R2FA	25A-IW-ST1-C-7029-01	1 OF 1	1	CJ-M15W13-PR-A20-000081	2	
33	647	R2FA	40A-IW-ST1-C-7003-01	1 OF 1	1	CJ-M15W14-PR-A20-000502	1A	
34	647	R2FA	50A-IW-ST1-E-7026-01	1 OF 1	0	CJ-M15W13-PR-A20-000081	2	
35	647	2FA	100A-IW-ST1-E-7025-01	1 OF 2	1	CJ-M15W13-PR-A20-000081	2	
36	647	2FA	100A-IW-ST1-E-7025-02	2 OF 2	1	CJ-M15W13-PR-A20-000081	2	
37	647	R2FA	100A-IW-ST1-E-7035-01	1 OF 1	0	CJ-M15W14-PR-A20-000507	1A	



NOTE
1. PUMP EXTERNAL SEAL FLUID LINES ON UDD TO BE DEVELOPED.
2. FOR FUTURE CONNECTION REF.

NO.	REVISION	DESCRIPTION	DATE
1	REVISED A/C	25.06.14	
2	REVISED A/C	25.01.17	
3	REVISED A/C	24.10.31	

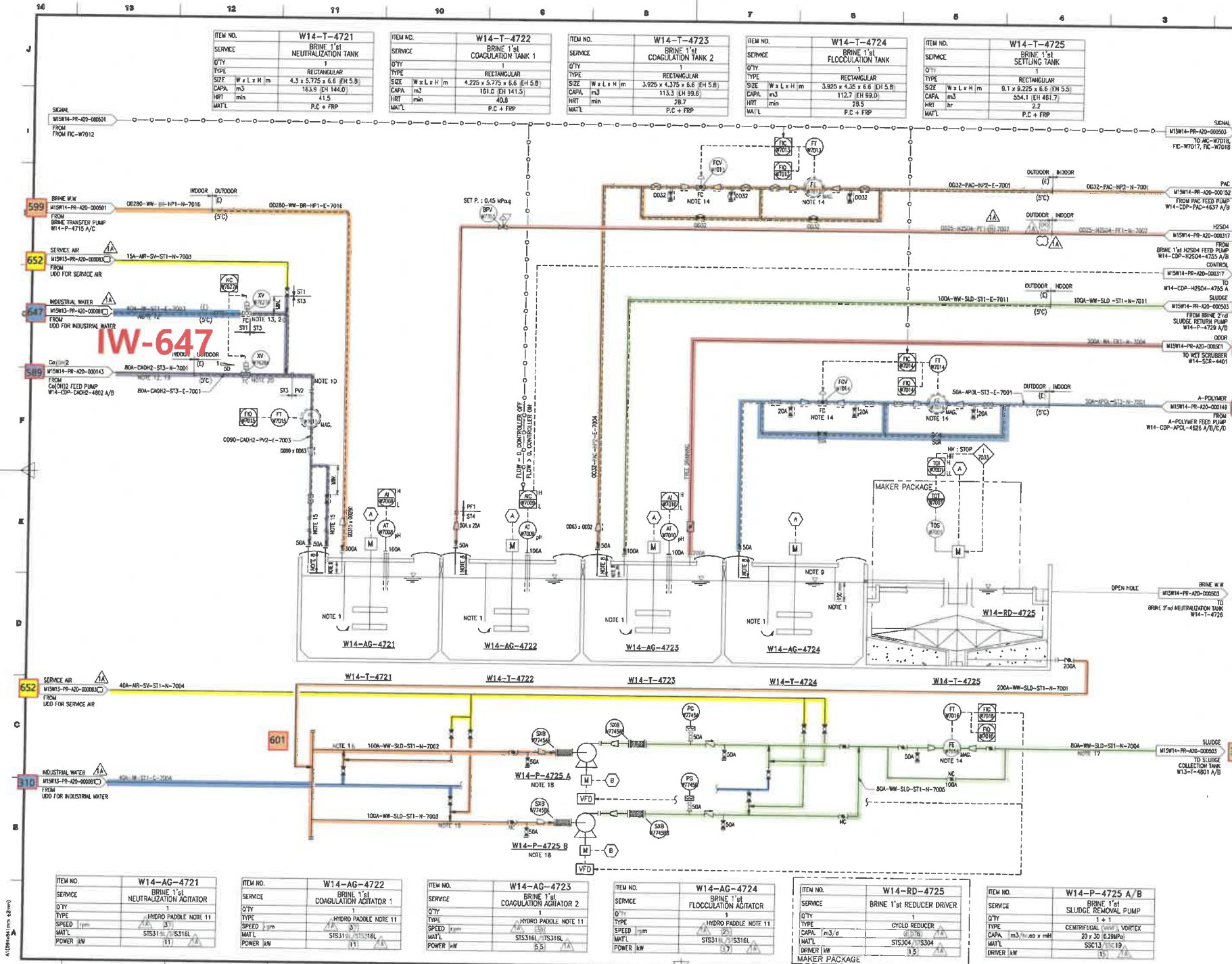
SK 에코플랜트	설계	이인혁
검토	한진희	
확인	박동영	
승인	박동영	

SK 에코플랜트
공사명
PROJECT TITLE
M15X Ph4 WWT Project

NO.	REVISION	DESCRIPTION	DATE
1	REVISED A/C	24.10.21	
2	REVISED A/C	24.02.25	
3	REVISED A/C	23.09.23	

SK 에코플랜트
공사명
PROJECT TITLE
M15X Ph3 WWT Project

UTILITY DISTRIBUTION DIAGRAM INDUSTRIAL WATER SUPPLY SYSTEM (1/3)	SCALE NONE	DATE 2024.10.31	PROJECT NO. 223300	REV 2
도면번호 SHEET NO. CJ-M15W13-PR-A20-000081				



대외비

KEY PLAN (1MF)

NOTE

1. SHUTTER VALVE: FRP
2. BATTLE DAMPER 0.02 m2 이상 AIR HOLE 설치
3. TANKER INHL 시비로 전방벽 모든 블록 BLOCK 번호 부여
4. MANHOLE: 2인 이상 설치, MANHOLE COVER: FRP
5. 3.2m 이상 높은 곳에 설치된 블록은 CHAIN VALVE 또는 PLATFORM 설치
6. MANHOLE: 2인 이상 설치, MANHOLE COVER: FRP
7. 3.2m 이상 높은 곳에 설치된 블록은 CHAIN VALVE 또는 PLATFORM 설치
8. CENTRIFUGAL PUMP, MATL: R4: CASING / IMPELLER
9. AGITATOR MATL: R4: SHAFT/BLADE
10. TANK IN 수위, 수위 제어 가능, 비파괴 수위 측정 (SLAB 조차까지 100mm 정밀도)
11. 4인 이상 VENDOR에 의함
12. 4인 이상 VENDOR에 의함
13. 4인 이상 VENDOR에 의함
14. INSTRUMENT VENDOR 연동: CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET SIZE SIZE 및 방향
15. 4인 이상 VENDOR에 의함
16. 4인 이상 VENDOR에 의함
17. SLUDGE REMOVAL PUMP 흡입 배관 PUMP SUCTION 4인 TANK 수위 HIGH POINT 설정
18. SLUDGE PUMP: TIMEON 시계 콘트롤
19. 2인 이상 VALVE HEADER에 있는 POINT 수위 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 시 MIN. DISTANCE 확보, 밸브 위치 및 수위 측정 가능
20. 4인 이상 VENDOR에 의함

FLUSHING WATER / AIR

MAIN FLOW

1A. REVISED A/C 25.08.14

1. REVISED A/C 25.01.17

0. APPROVAL FOR CONSTRUCTION 24.10.01

수명: 10년

설계: SK E&E

검核: SK E&E

제작: SK E&E

설치: SK E&E

유지: SK E&E

SK E&E

공 사 명

PROJECT TITLE

M15X Ph4 WMT Project

도면명

OWN TITLE

PIPING & INSTRUMENT DIAGRAM

BRINE 1st COAGULATION & SETTLING TANK

속 치

SCALE

NONE

날

DATE

2025. 08. 14

PROJECT NO.

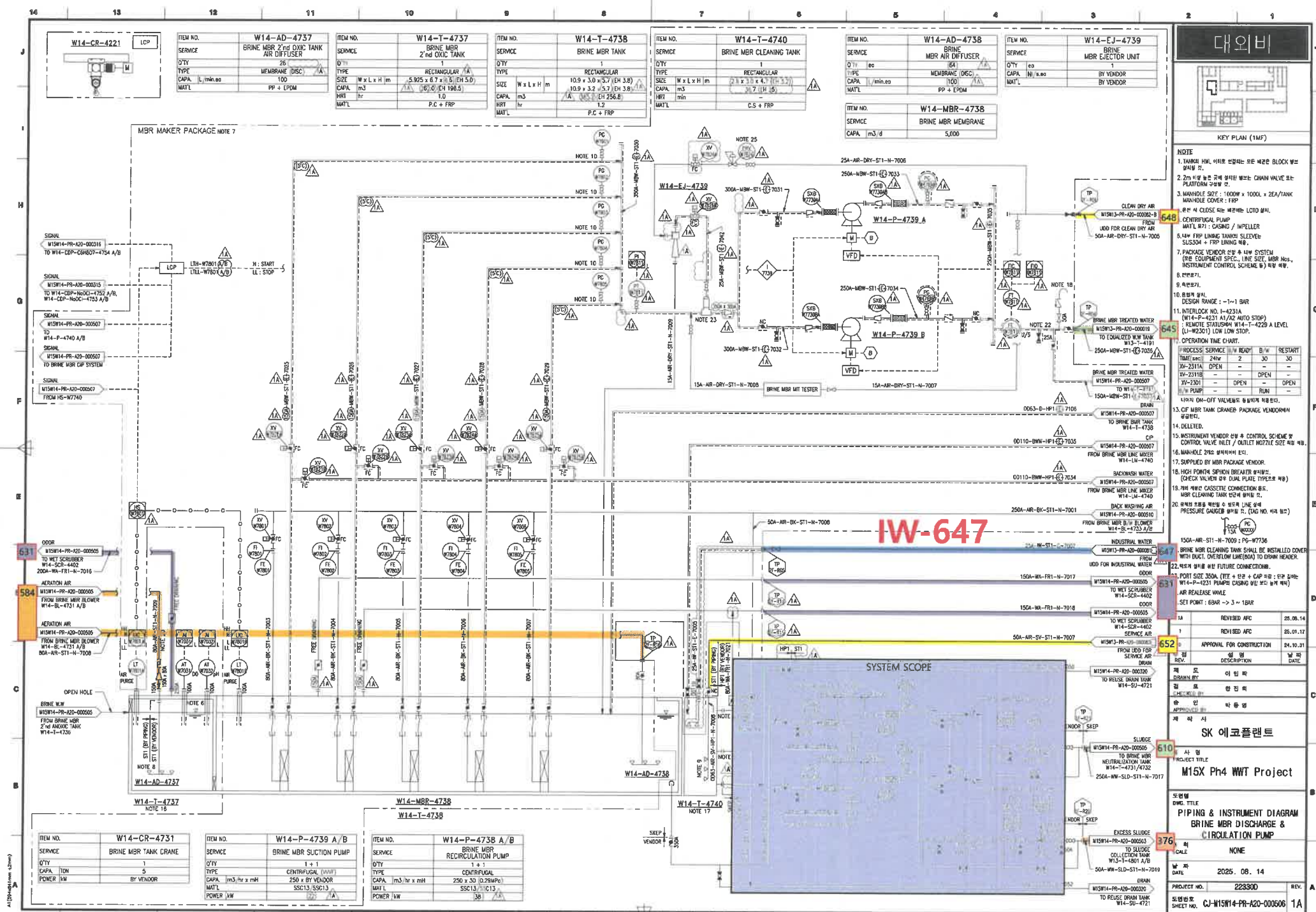
223300

도면번호

SHEET NO.

CJ-M15W14-PR-A20-000502

1A



대외비

KEY PLAN (1MF)

NOTE

- TANKER H/WL 이하로 설치되는 모든 배관 BLOCK은 배관 R.
- 2m 이상 높음 구역 설치되는 CHAIN WALK PLATFORM 2중 2.
- MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP
- 호수 시 CLOSE 하는 배관은 LOTO 할.
- CENTRIFUGAL PUMP MAT'L B7 : CASING / IMPELLER SUS304 + TPO LINING MB.
- 5.4W FIBER LINING TANK SLEEVE SUS304 + TPO LINING MB.
- PACKAGE VENDOR 장비 & M/W SYSTEM (P&ID EQUIPMENT SPEC., LINE SIZE, MBR H/W, INSTRUMENT CONTROL SCHEME 등) 적용 함.
- 9.4.2021.
- 8.4.2021.
- 10.8.2021.
- INTERLOCK NO. 1-4231A (W14-P-4231 A1/A2 AUTO STOP)
- RELIEF STATUS: W14-T-4239 A LEVEL (L1-W2201) LOW LOW STOP.
- OPERATION TIME CHART:

PROCESS	SERVICE	READY	B/W	RESTART
W14-211A	OPEN	2	30	30
W14-211B	OPEN	2	30	30
W14-211C	OPEN	2	30	30
W14-211D	OPEN	2	30	30
W14-211E	OPEN	2	30	30
W14-211F	OPEN	2	30	30
W14-211G	OPEN	2	30	30
W14-211H	OPEN	2	30	30
W14-211I	OPEN	2	30	30
W14-211J	OPEN	2	30	30
W14-211K	OPEN	2	30	30
W14-211L	OPEN	2	30	30
W14-211M	OPEN	2	30	30
W14-211N	OPEN	2	30	30
W14-211O	OPEN	2	30	30
W14-211P	OPEN	2	30	30
W14-211Q	OPEN	2	30	30
W14-211R	OPEN	2	30	30
W14-211S	OPEN	2	30	30
W14-211T	OPEN	2	30	30
W14-211U	OPEN	2	30	30
W14-211V	OPEN	2	30	30
W14-211W	OPEN	2	30	30
W14-211X	OPEN	2	30	30
W14-211Y	OPEN	2	30	30
W14-211Z	OPEN	2	30	30

13. OF MBR TANK CRANE: PACKAGE VENDOR에 의함.

14. DELETED.

15. INSTRUMENT VENDOR 장비 & CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 적용 함.

16. MANHOLE 2중 설치되야 함.

17. SUPPLIED BY MBR PACKAGE VENDOR.

18. HIGH POINT SIPHON BREAKER #111111 (CHECK VALVE & DUAL PLATE TYPE 적용 함)

19. H/W PUMP CASSETTE CONNECTION B/E. MBR CLEANING TANK C/SCH #111111.

20. 유량계 흐름 방향을 수 표시 LINE 상에 PRESSURE GAUGE #111111 (TAG NO. 111111)

150A-AIR-ST1-N-7009 : PG-W736

BRINE MBR CLEANING TANK SHALL BE INSTALLED DOWN WITH DRAIN LINE CONNECTED TO DRAIN HEADER.

22. 배관 방향 및 FUTURE CONNECTION.

PORT SIZE 350A (H/W + H/W + CAP #111111 : 111111) AIR RELEASE VALVE

SET POINT : 6BAR -> 3 - 1BAR

REVISED AFC 25.08.14

REVISED AFC 25.01.17

APPROVAL FOR CONSTRUCTION 24.10.31

SK 에코플랜트

사명

M15X Ph4 WWT Project

도면명

PIPING & INSTRUMENT DIAGRAM

BRINE MBR DISCHARGE & CIRCULATION PUMP

SCALE

NONE

날짜

2025. 08. 14

PROJECT NO.

223300

REV.

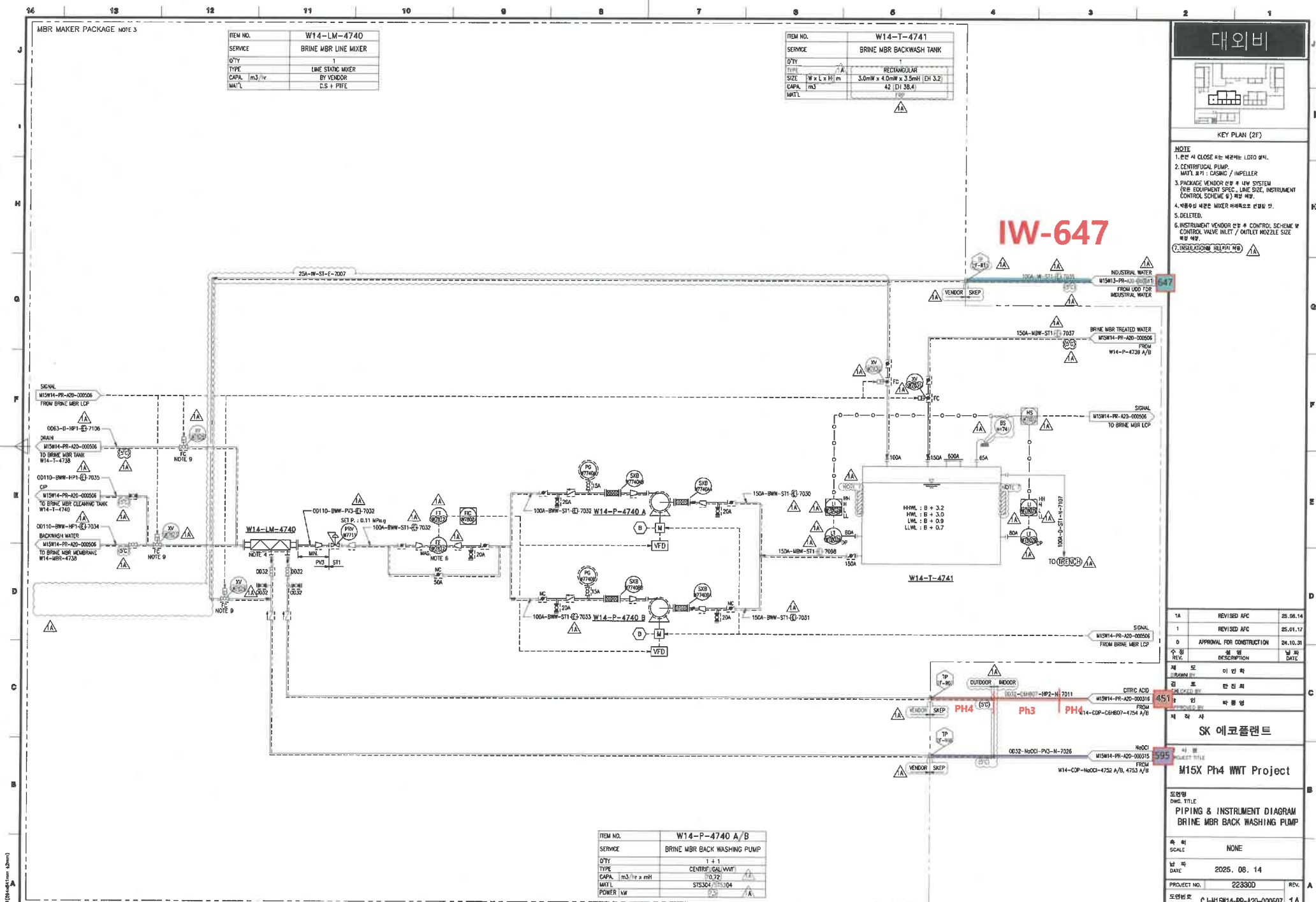
A

도면번호

223300

시트명

CJ-W15W14-PR-A20-000506 1A



ITEM NO.	W14-LM-4740
SERVICE	BRINE MBR LINE MIXER
QTY	1
TYPE	LINE STATIC MIXER
CAPA. m3/hr	BY VENDOR
MAT'L	C.S. + PTFE

ITEM NO.	W14-T-4741
SERVICE	BRINE MBR BACKWASH TANK
QTY	1
TYPE	RECTANGULAR
SIZE W x L x H m	3.0mW x 4.0mL x 3.5mH (EH 3.2)
CAPA. m3	42 (DI 38.4)
MAT'L	SSP

ITEM NO.	W14-P-4740 A/B
SERVICE	BRINE MBR BACK WASHING PUMP
QTY	1 + 1
TYPE	CENTRIF. GALV/WF
CAPA. m3/hr x mH	70.72
MAT'L	STS304
POWER kW	22

대외비

KEY PLAN (2F)

NOTE

1. 관련 시 CLOSE 시는 배관내 LOTO 하기.
2. CENTRIFUGAL PUMP.
MAT'L WP1 : SS304 / IMPELLER
3. PACKAGE VENDOR 사양 및 LWF SYSTEM
(SHEET EQUIPMENT SPEC., LINE SIZE, INSTRUMENT
CONTROL SCHEME 등) 확인 하기.
4. 역송수압 때문에 MIXER HORIZONTALLY 설치함.
5. DELETED.
6. INSTRUMENT VENDOR 사양 & CONTROL SCHEME W
CONTROL VALVE INLET / OUTLET NOZZLE SIZE
확인 하기.
7. INSULATION HIGHLIGHT 하기

1A	REVISED APC	25.08.14
1	REVISED APC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31

수정	설명	날짜
REV.	DESCRIPTION	DATE
제	이	인
DRWN	BY	
검	토	한
CHK	BY	
승	인	박
APPR	BY	
제	각	사

SK 에코플랜트

M15X Ph4 WWT Project

도면명
OWN TITLE
PIPING & INSTRUMENT DIAGRAM
BRINE MBR BACK WASHING PUMP

속 력
SCALE
NONE

날 짜
DATE
2025. 08. 14

PROJECT NO.
223300

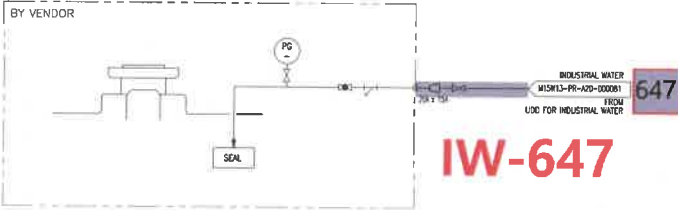
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SHEET NO.
CJ-W15W14-PR-A20-000507

REV.
1A

대외비

NOTE

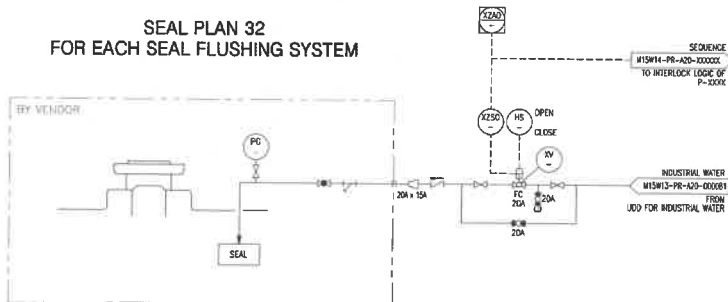
SEAL PLAN 32 FOR EACH SEAL FLUSHING SYSTEM



ITEM NO.	P&ID NO.	PG	EXTERNAL SEAL FLUID LINE NO.
W14-P-4106 A1	M15W14-PR-A20-000102	PG-WB156A1	20A-W-ST1-C-9741
W14-P-4106 B2	M15W14-PR-A20-000102	PG-WB156A2	20A-W-ST1-C-9742
W14-P-4110 A1	M15W14-PR-A20-000103	PG-WB156A1	20A-W-ST1-C-9743
W14-P-4110 B2	M15W14-PR-A20-000103	PG-WB156A2	20A-W-ST1-C-9744
W14-P-4106 B1	M15W14-PR-A20-000104	PG-WB156B1	20A-W-ST1-C-9745
W14-P-4106 B2	M15W14-PR-A20-000104	PG-WB156B2	20A-W-ST1-C-9746
W14-P-4110 B1	M15W14-PR-A20-000105	PG-WB160B1	20A-W-ST1-C-9747
W14-P-4110 B2	M15W14-PR-A20-000105	PG-WB160B2	20A-W-ST1-C-9748
W14-P-4121 A1	M15W14-PR-A20-000106-A	PG-WB171A1	20A-W-ST1-C-9701
W14-P-4121 B1	M15W14-PR-A20-000106-A	PG-WB171B1	20A-W-ST1-C-9702
W14-P-4121 A2	M15W14-PR-A20-000107	PG-WB176A1	20A-W-ST1-C-9749
W14-P-4121 B2	M15W14-PR-A20-000107	PG-WB176A2	20A-W-ST1-C-9750
W14-P-4130 A1	M15W14-PR-A20-000108	PG-WB180A1	20A-W-ST1-C-9751
W14-P-4130 A2	M15W14-PR-A20-000108	PG-WB180A2	20A-W-ST1-C-9752
W14-P-4126 B1	M15W14-PR-A20-000109	PG-WB176B1	20A-W-ST1-C-9753
W14-P-4126 B2	M15W14-PR-A20-000109	PG-WB176B2	20A-W-ST1-C-9754
W14-P-4130 B1	M15W14-PR-A20-000110	PG-WB180B1	20A-W-ST1-C-9755
W14-P-4130 B2	M15W14-PR-A20-000110	PG-WB180B2	20A-W-ST1-C-9756
W14-P-4221 A1	M15W14-PR-A20-000111	PG-WB271A1	20A-W-ST1-C-9705
W14-P-4221 B1	M15W14-PR-A20-000111	PG-WB271B1	20A-W-ST1-C-9706
W14-P-4221 A2	M15W14-PR-A20-000112	PG-WB276A1	20A-W-ST1-C-9801
W14-P-4221 B2	M15W14-PR-A20-000112	PG-WB276A2	20A-W-ST1-C-9802
W14-P-4230 A1	M15W14-PR-A20-000113	PG-WB280A1	20A-W-ST1-C-9710
W14-P-4230 A2	M15W14-PR-A20-000113	PG-WB280A2	20A-W-ST1-C-9711
W14-P-4225 B1	M15W14-PR-A20-000114	PG-WB276B1	20A-W-ST1-C-9803
W14-P-4225 B2	M15W14-PR-A20-000114	PG-WB276B2	20A-W-ST1-C-9804

ITEM NO.	P&ID NO.	PG	EXTERNAL SEAL FLUID LINE NO.
W14-P-4230 B1	M15W14-PR-A20-000115	PG-WB280B1	20A-W-ST1-C-9714
W14-P-4230 B2	M15W14-PR-A20-000115	PG-WB280B2	20A-W-ST1-C-9715
W14-P-4191 A1	M15W14-PR-A20-000119	PG-WB241A1	20A-W-ST1-C-9716
W14-P-4191 B1	M15W14-PR-A20-000119	PG-WB241B1	20A-W-ST1-C-9717
W14-P-4191 C1	M15W14-PR-A20-000119	PG-WB241C1	20A-W-ST1-C-9718
W14-P-4195 A1	M15W14-PR-A20-000120	PG-WB245A1	20A-W-ST1-C-9719
W14-P-4195 A2	M15W14-PR-A20-000120	PG-WB245A2	20A-W-ST1-C-9720
W14-P-4195 B1	M15W14-PR-A20-000121	PG-WB245B1	20A-W-ST1-C-9721
W14-P-4195 B2	M15W14-PR-A20-000121	PG-WB245B2	20A-W-ST1-C-9722
W14-P-4801 A1	M15W14-PR-A20-000134	PG-WB551A1	20A-W-ST1-C-9727
W14-P-4801 B1	M15W14-PR-A20-000134	PG-WB551B1	20A-W-ST1-C-9728
W14-P-4807 A1	M15W14-PR-A20-000163	PG-WB557A1	20A-W-ST1-C-9729
W14-P-4807 B1	M15W14-PR-A20-000163	PG-WB557B1	20A-W-ST1-C-9730
W14-P-4715 A1	M15W14-PR-A20-000501	PG-WB715A1	20A-W-ST1-C-7008
W14-P-4715 B1	M15W14-PR-A20-000501	PG-WB715B1	20A-W-ST1-C-7009
W14-P-4725 A1	M15W14-PR-A20-000502	PG-WB725A1	20A-W-ST1-C-7011
W14-P-4725 B1	M15W14-PR-A20-000502	PG-WB725B1	20A-W-ST1-C-7012
W14-P-4729 A1	M15W14-PR-A20-000503	PG-WB729A1	20A-W-ST1-C-7013
W14-P-4729 B1	M15W14-PR-A20-000503	PG-WB729B1	20A-W-ST1-C-7014
W14-P-4730 A1	M15W14-PR-A20-000504	PG-WB730A1	20A-W-ST1-C-7015
W14-P-4730 B1	M15W14-PR-A20-000504	PG-WB730B1	20A-W-ST1-C-7016
W14-P-4735 A1	M15W14-PR-A20-000505	PG-WB735A1	20A-W-ST1-C-7017
W14-P-4735 B1	M15W14-PR-A20-000505	PG-WB735B1	20A-W-ST1-C-7018
W14-P-4738 A1	M15W14-PR-A20-000506	PG-WB738A1	20A-W-ST1-C-7019
W14-P-4738 B1	M15W14-PR-A20-000506	PG-WB738B1	20A-W-ST1-C-7020
W14-P-4731 A1/A2	M15W14-PR-A20-000113		
W14-P-4731 B1/B2	M15W14-PR-A20-000113		
W14-P-4232 A/B	M15W14-PR-A20-000116		
W14-P-4739 A/B	M15W14-PR-A20-000506		
W14-P-4191 A/B	M15W14-PR-A20-000119		

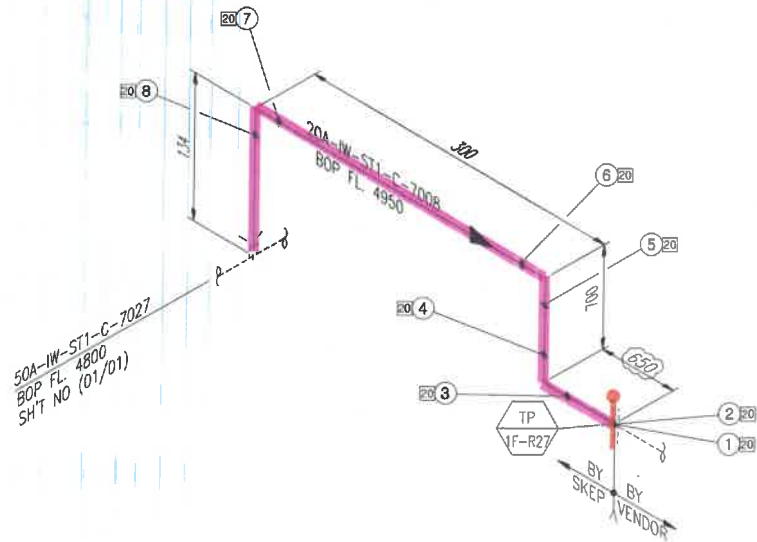
SEAL PLAN 32 FOR EACH SEAL FLUSHING SYSTEM



ITEM NO.	P&ID NO.	PG	XV	HS	XZSD	XZAD	EXTERNAL SEAL FLUID LINE NO.
W14-P-4805 A1	M15W14-PR-A20-000135	PG-WB855A1	XV-WB855A1	HS-WB855A1	XZSD-WB855A1	XZAD-WB855A1	20A-W-ST1-C-9737
W14-P-4805 A2	M15W14-PR-A20-000135	PG-WB855A2	XV-WB855A2	HS-WB855A2	XZSD-WB855A2	XZAD-WB855A2	20A-W-ST1-C-9738
W14-P-4805 B1	M15W14-PR-A20-000136	PG-WB855B1	XV-WB855B1	HS-WB855B1	XZSD-WB855B1	XZAD-WB855B1	20A-W-ST1-C-9739
W14-P-4805 B2	M15W14-PR-A20-000136	PG-WB855B2	XV-WB855B2	HS-WB855B2	XZSD-WB855B2	XZAD-WB855B2	20A-W-ST1-C-9740

28	REVISED AFC	25.08.14
2	REVISED AFC	28.01.17
1	REVISED AFC	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE
검 CHECK BY	이 이	한 한
검 CHECK BY	한 한	진 진
승 APPROVED BY	인 인	박 박
제 사 사	SK	에코플랜트
공 PROJECT TITLE	M15X Ph4 WWT Project	
도 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM PUMP SEAL & PIPING PLAN (1/2)	
축 SCALE	NONE	
날 DATE	2025. 08. 14	
PROJ. NO.	223300	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000011	2A

11



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CONVED : RC * EDC RED : RE * CON SWAGE : RCS/HCS1 * EDC SWAGE : RES/HES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HOPE/PMC) : SC * SADDLE(HOPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VA/VA1/VA2 * DIAPHRAGM VA : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(SHIELD) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(I-TYPE) : YSTR * STRAINER(Z-TYPE) : ZSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : IR/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	RIFA	REF. PLAN DWG.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	25	DESIGN	13.5	60					CJ-M15W14-PI-A30-010017-B	25.03.31	REVISED AS MARKED	왕래훈	한남주	박동영
			OPERATING	10	AMB					CJ-M15W13-PR-A20-000081	25.01.24	APPROVED FOR CONSTRUCTION	왕래훈	한남주	박동영

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	20A	1.8M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	20A	3EA	ELBOW 90 DEG, STS304, SW - SCH 10S

F 20A 1EA FLG SO, KS 10K, RF, STS304

IN 25X20A 1.8M 관 보온재(고무발포)

CLIENT

SK hynix

1

CONTRACTOR

SK ecoplant

PROJECT NAME

M15X Ph4 Project

COMMODITY

IW

JOB NO.

22330D

LINE NO.

20A-IW-ST1-C-7008

SHT NO.

REV.

01 / 01

1



CLIENT		3	
			
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.		SH'T NO.	REV.
20A-IW-ST1-C-7010		01 / 01	1

[illegible]

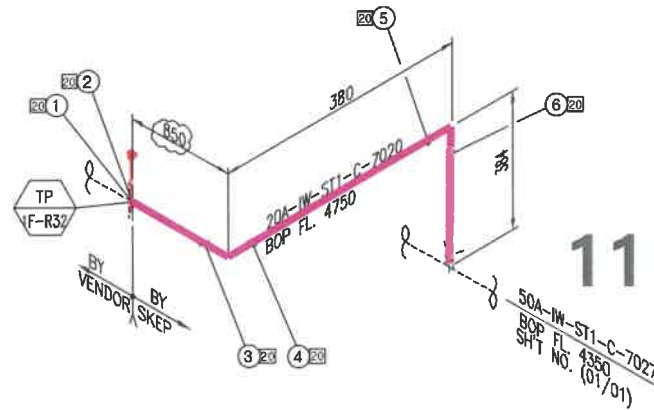
경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



* PIPE: P/P1	* 90° ELBOW: E90	* 45° ELBOW: E45	* TEE: T	* RED TEE: TR	* STUB-IN: STU	* REINFORCING PAD: RP	* COULRED: RC	* ECC RED: RE	* CON SNAKE: RCS/RCS1	* ECC SNAKE: RES/RES1	* NIPPLE: NP/NP1/NP2	* CAP: CAP/CAP1	* PLUG: PLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* STUB-IN: SU	* SOCKET/HALF PAD: SC	* SHOULDER/HALF: SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* BLIND FLANGE: FB/FB1	* GATE VA: VG/G1/G2	* BALL VA: VG/G1/G2	* BUTTERFLY VA: VB/VB1	* CHECK VA: VC/CA1/VC2/CA2/VC3/CA3/VC4/CA4/VC5/CA5					
* GLOBE VA: VA/VA1/VA2	* DIAPHRAGM VA: VD/VD1	* DAMPER: VP	* QUICK COUPLING: SC	* FLEXIBLE JOINT(STEE): FLXJ	* FLEXIBLE JOINT(RUBBER): FLXR	* FLEXIBLE JOINT(STEE PLTE LINED): FLXL	* FLEXIBLE HOSE: FHS								
* STRAINER(Y-TYPE): YSTR	* STRAINER(T-TYPE): TSTR	* STRAINER(BLOCK-TYPE): BSTR	* QSC	* ANCHOR PLATE: STOP	* BIRD SCREEN: BS	* ISOLATION KIT: IK	* UNLOADING BOX: UB	* AIR TAP: AT	* LOTO: LOTO	* SIGHT GLASS: SG	* PRESSURE GAUGE: PG	* SHUT: SH	* SPECIAL HUNTER: SH1/SH11		

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14								CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	13.5	60	UNIT RIFA									M15X Ph4 Project
			OPERATING	10	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-D1D017-B		25.03.31	REVISED AS MARKED	황래중	안남후	박동영		IW	22330D
						REF. P & ID		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		LINE NO.	SH'T NO.
						CJ-M15W13-PR-A20-000081	DATE	DESCRIPTION	BY	CHK	APP'D		20A-IW-ST1-C-7015	.01 / .01	REV. 1

경고: 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음.

[illegible]

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* REDUCE : TR	* SUB-IN : SU	* REINFORCING PAD : RP	* CONE/RED : RC	* EGG/RED : RE	* CON/SNAGE : RCS/RCS1	* EGG/SNAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* SUB-IN : SU	* SOCKET(NIPPLE/PVC) : SG	* SOCKET(NIPPLE) : SO	* FLANGE : F/F1	* FLASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB2/VB3/VB4/VB5/VB6/VB7/VB8/VB9/VB10/VB11/VB12/VB13/VB14/VB15/VB16/VB17/VB18/VB19/VB20/VB21/VB22/VB23/VB24/VB25/VB26/VB27/VB28/VB29/VB30/VB31/VB32/VB33/VB34/VB35/VB36/VB37/VB38/VB39/VB40/VB41/VB42/VB43/VB44/VB45/VB46/VB47/VB48/VB49/VB50/VB51/VB52/VB53/VB54/VB55/VB56/VB57/VB58/VB59/VB60/VB61/VB62/VB63/VB64/VB65/VB66/VB67/VB68/VB69/VB70/VB71/VB72/VB73/VB74/VB75/VB76/VB77/VB78/VB79/VB80/VB81/VB82/VB83/VB84/VB85/VB86/VB87/VB88/VB89/VB90/VB91/VB92/VB93/VB94/VB95/VB96/VB97/VB98/VB99/VB100/VB101/VB102/VB103/VB104/VB105/VB106/VB107/VB108/VB109/VB110/VB111/VB112/VB113/VB114/VB115/VB116/VB117/VB118/VB119/VB120/VB121/VB122/VB123/VB124/VB125/VB126/VB127/VB128/VB129/VB130/VB131/VB132/VB133/VB134/VB135/VB136/VB137/VB138/VB139/VB140/VB141/VB142/VB143/VB144/VB145/VB146/VB147/VB148/VB149/VB150/VB151/VB152/VB153/VB154/VB155/VB156/VB157/VB158/VB159/VB160/VB161/VB162/VB163/VB164/VB165/VB166/VB167/VB168/VB169/VB170/VB171/VB172/VB173/VB174/VB175/VB176/VB177/VB178/VB179/VB180/VB181/VB182/VB183/VB184/VB185/VB186/VB187/VB188/VB189/VB190/VB191/VB192/VB193/VB194/VB195/VB196/VB197/VB198/VB199/VB200/VB201/VB202/VB203/VB204/VB205/VB206/VB207/VB208/VB209/VB210/VB211/VB212/VB213/VB214/VB215/VB216/VB217/VB218/VB219/VB220/VB221/VB222/VB223/VB224/VB225/VB226/VB227/VB228/VB229/VB230/VB231/VB232/VB233/VB234/VB235/VB236/VB237/VB238/VB239/VB240/VB241/VB242/VB243/VB244/VB245/VB246/VB247/VB248/VB249/VB250/VB251/VB252/VB253/VB254/VB255/VB256/VB257/VB258/VB259/VB260/VB261/VB262/VB263/VB264/VB265/VB266/VB267/VB268/VB269/VB270/VB271/VB272/VB273/VB274/VB275/VB276/VB277/VB278/VB279/VB280/VB281/VB282/VB283/VB284/VB285/VB286/VB287/VB288/VB289/VB290/VB291/VB292/VB293/VB294/VB295/VB296/VB297/VB298/VB299/VB300/VB301/VB302/VB303/VB304/VB305/VB306/VB307/VB308/VB309/VB310/VB311/VB312/VB313/VB314/VB315/VB316/VB317/VB318/VB319/VB320/VB321/VB322/VB323/VB324/VB325/VB326/VB327/VB328/VB329/VB330/VB331/VB332/VB333/VB334/VB335/VB336/VB337/VB338/VB339/VB340/VB341/VB342/VB343/VB344/VB345/VB346/VB347/VB348/VB349/VB350/VB351/VB352/VB353/VB354/VB355/VB356/VB357/VB358/VB359/VB360/VB361/VB362/VB363/VB364/VB365/VB366/VB367/VB368/VB369/VB370/VB371/VB372/VB373/VB374/VB375/VB376/VB377/VB378/VB379/VB380/VB381/VB382/VB383/VB384/VB385/VB386/VB387/VB388/VB389/VB390/VB391/VB392/VB393/VB394/VB395/VB396/VB397/VB398/VB399/VB400/VB401/VB402/VB403/VB404/VB405/VB406/VB407/VB408/VB409/VB410/VB411/VB412/VB413/VB414/VB415/VB416/VB417/VB418/VB419/VB420/VB421/VB422/VB423/VB424/VB425/VB426/VB427/VB428/VB429/VB430/VB431/VB432/VB433/VB434/VB435/VB436/VB437/VB438/VB439/VB440/VB441/VB442/VB443/VB444/VB445/VB446/VB447/VB448/VB449/VB450/VB451/VB452/VB453/VB454/VB455/VB456/VB457/VB458/VB459/VB460/VB461/VB462/VB463/VB464/VB465/VB466/VB467/VB468/VB469/VB470/VB471/VB472/VB473/VB474/VB475/VB476/VB477/VB478/VB479/VB480/VB481/VB482/VB483/VB484/VB485/VB486/VB487/VB488/VB489/VB490/VB491/VB492/VB493/VB494/VB495/VB496/VB497/VB498/VB499/VB500/VB501/VB502/VB503/VB504/VB505/VB506/VB507/VB508/VB509/VB510/VB511/VB512/VB513/VB514/VB515/VB516/VB517/VB518/VB519/VB520/VB521/VB522/VB523/VB524/VB525/VB526/VB527/VB528/VB529/VB530/VB531/VB532/VB533/VB534/VB535/VB536/VB537/VB538/VB539/VB540/VB541/VB542/VB543/VB544/VB545/VB546/VB547/VB548/VB549/VB550/VB551/VB552/VB553/VB554/VB555/VB556/VB557/VB558/VB559/VB560/VB561/VB562/VB563/VB564/VB565/VB566/VB567/VB568/VB569/VB570/VB571/VB572/VB573/VB574/VB575/VB576/VB577/VB578/VB579/VB580/VB581/VB582/VB583/VB584/VB585/VB586/VB587/VB588/VB589/VB590/VB591/VB592/VB593/VB594/VB595/VB596/VB597/VB598/VB599/VB600/VB601/VB602/VB603/VB604/VB605/VB606/VB607/VB608/VB609/VB610/VB611/VB612/VB613/VB614/VB615/VB616/VB617/VB618/VB619/VB620/VB621/VB622/VB623/VB624/VB625/VB626/VB627/VB628/VB629/VB630/VB631/VB632/VB633/VB634/VB635/VB636/VB637/VB638/VB639/VB640/VB641/VB642/VB643/VB644/VB645/VB646/VB647/VB648/VB649/VB650/VB651/VB652/VB653/VB654/VB655/VB656/VB657/VB658/VB659/VB660/VB661/VB662/VB663/VB664/VB665/VB666/VB667/VB668/VB669/VB670/VB671/VB672/VB673/VB674/VB675/VB676/VB677/VB678/VB679/VB680/VB681/VB682/VB683/VB684/VB685/VB686/VB687/VB688/VB689/VB690/VB691/VB692/VB693/VB694/VB695/VB696/VB697/VB698/VB699/VB700/VB701/VB702/VB703/VB704/VB705/VB706/VB707/VB708/VB709/VB710/VB711/VB712/VB713/VB714/VB715/VB716/VB717/VB718/VB719/VB720/VB721/VB722/VB723/VB724/VB725/VB726/VB727/VB728/VB729/VB730/VB731/VB732/VB733/VB734/VB735/VB736/VB737/VB738/VB739/VB740/VB741/VB742/VB743/VB744/VB745/VB746/VB747/VB748/VB749/VB750/VB751/VB752/VB753/VB754/VB755/VB756/VB757/VB758/VB759/VB760/VB761/VB762/VB763/VB764/VB765/VB766/VB767/VB768/VB769/VB770/VB771							

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	13.5	60	UNIT R1FA								M15X Ph4 Project
			OPERATING	10	AMB	REF. PLAN DWG.							COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-010016-B		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	IW	22330D
						REF. P & ID		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	SH'T NO.
						CJ-M15W13-PR-A20-000081	REV. DATE	DESCRIPTION	BY	CHK	APP'D	20A-W-S1-C-7020	01 / 01	REV. 1

CLIENT



7

CONTRACTOR	
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02	COMMODITY
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14

영	LINE NO.
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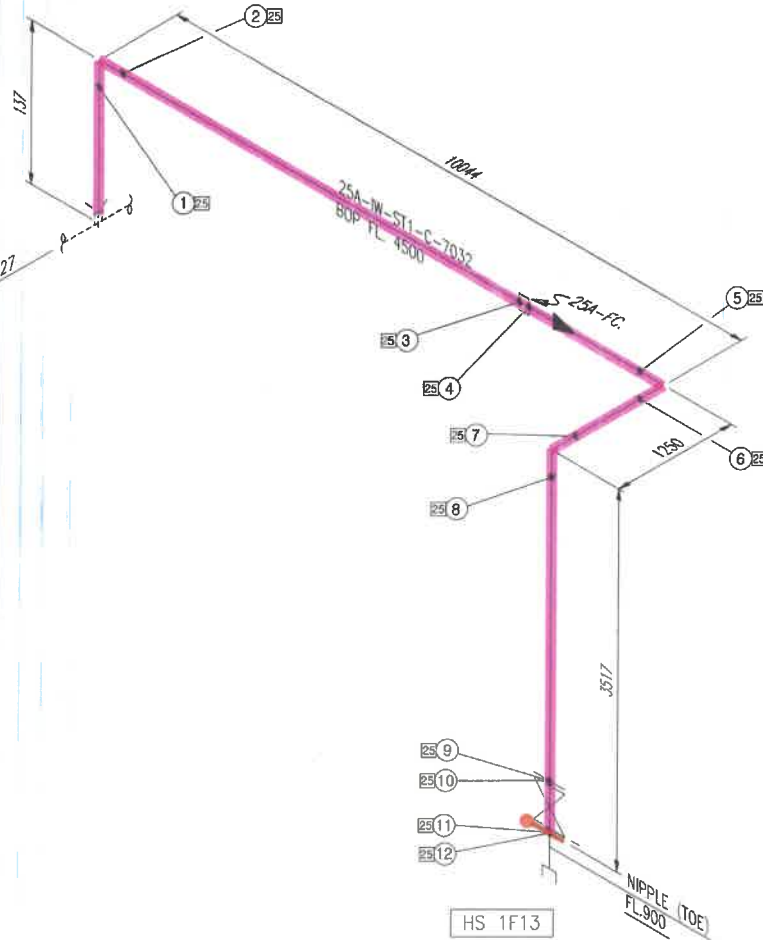
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PROJECT NAME	M15X Ph4 Project
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JOB NO. 22330D

SH'T NO.	REV.
01 / 01	1

11

50A-IW-ST1-C-7027
BOP FL. 4350
SH'T NO (01/01)

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CONFLUENT : RC * ECC SWAGE : RES/RS1 * ECC SWAGE : RES/RS2 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PMC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VAL : VG/VG1/VG2 * BALL VAL : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VAL : VF/VF1 * CHECK VAL : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VAL : VL/VL1/VL2 * DIAPHRAGM VAL : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(I-TYPE) : ISTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R1FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-B	REF. P & ID	CJ-M15W13-PR-A20-000081	REV.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	25	DESIGN	13.5	60														
			OPERATING	10	AMB														

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	25A	14.9M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	3EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
FC	25A	1EA	COUPLING FULL, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLGD, BODY-STSF304 / TRIM-STSF304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	25A	2SET	MACHINE BOLT & NUTS : STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)
IN	25IX25A	14.9M	관 보온재(2중발포)

CLIENT

SK hynix

8

CONTRACTOR

SK ecoplant

PROJECT NAME

M15X Ph4 Project

COMMODITY

IW

JOB NO.

22390D

LINE NO.

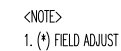
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SH'T NO.

01 / 01

REV.

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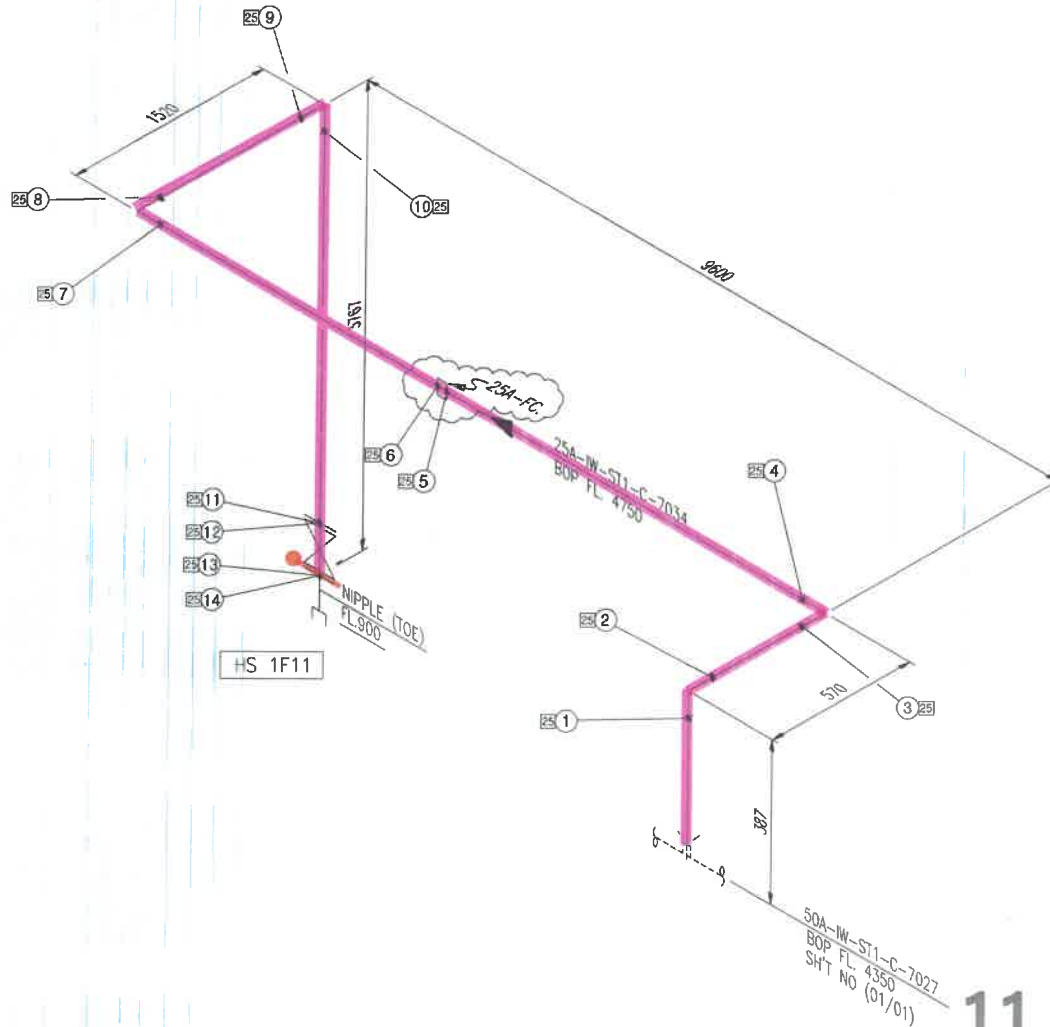


CLIENT  9

CONTRACTOR 		PROJECT NAME	
COMMODITY		JOB NO.	
IW		23330D	
LINE NO.	25A-IW-ST1-C-7033	SH'T NO. 01 / 01	REV. 0

* PIPE: P/P1	* 90° ELBOW: E90	* 45° ELBOW: E45	* TEE: T	* RED.TEE: TR	* STUB-IN: STU	* REINFORCING PAD: RP	* CON.RED: RC	* ECC.RED: RE	* CON.SWAGE: RCS/RCS1	* ECC.SWAGE: RES/RES1	* NP/PLE: NP/NP1/NP2	* CAP: CAP/CAP1	* PLUG: PLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* STUB-IN: STU	* SOCKET(HOPE/PWC): SC	* SADDLE(HOPE): SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* BLIND FLANGE: FB/FB1	* GATE VA: VG/VG1/VG2	* BALL VA: VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA: VF/VF4	* CHECK VA: VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA: VL/VL1/VL2	* DIAPHRAGM VA: VD/VD0	* DAMPER: VP	* QUICK COUPLING: SQC	* FLEXIBLE JOINT(STEEL): FLXB	* FLEXIBLE JOINT(RUBBER): FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED): FLXL	* FLEXIBLE HOSE: FLXH	* STEAM TRAP: ST	* SPRAY NOZZLE: SPN	* EMERGENCY SHOWER: SSH					
* STRAINER(Y-TYPE): YSTR	* STRAINER(T-TYPE): TSTR	* STRAINER(BUCKET-TYPE): BSTR	* ANCHOR PLATE: STOP	* BIRD SCREEN: BS	* ISOLATION KIT: IK	* UNLOADING BOX: UB	* AIR TRAP: AT	* LOTO: LOTO	* SIGHT GLASS: SG	* PRESSURE GAUGE: PG	* SHILED: SH	* SPECIAL BOLT & NUT: B/B1			

[illegible]



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	25A	15.8M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
FC	25A	1EA	COUPLING FULL, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 / TRIM-STSF304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	25A	2SET	MACHINE BOLT & NUTS STS 304 / W WASHERS
SOC	25A	1EA	QUICK COUPLING(MALE)
IN	25X25A	15.8M	관 보편(24인치)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * SUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RES1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
 * SUB-IN : STU * SOCKET(HDPE/PMG) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/YB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB110/VB2/VB210 * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VCS/VCS/VCS * GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VMD * DAMPER : VP * QUICK COUPLING : SOC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
 * STRAINER(PY-TYPE) : YSTR * STRAINER(I-TYPE) : ISTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R1FA	REF. PLAN DWG.	REF. P & ID	REV	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	25	DESIGN	13.5	60					CJ-M15W14-PI-A30-010016-B	CJ-M15W13-PR-A20-000081	0	25.01.24	APPROVED FOR CONSTRUCTION	황래경	안남루	박동영

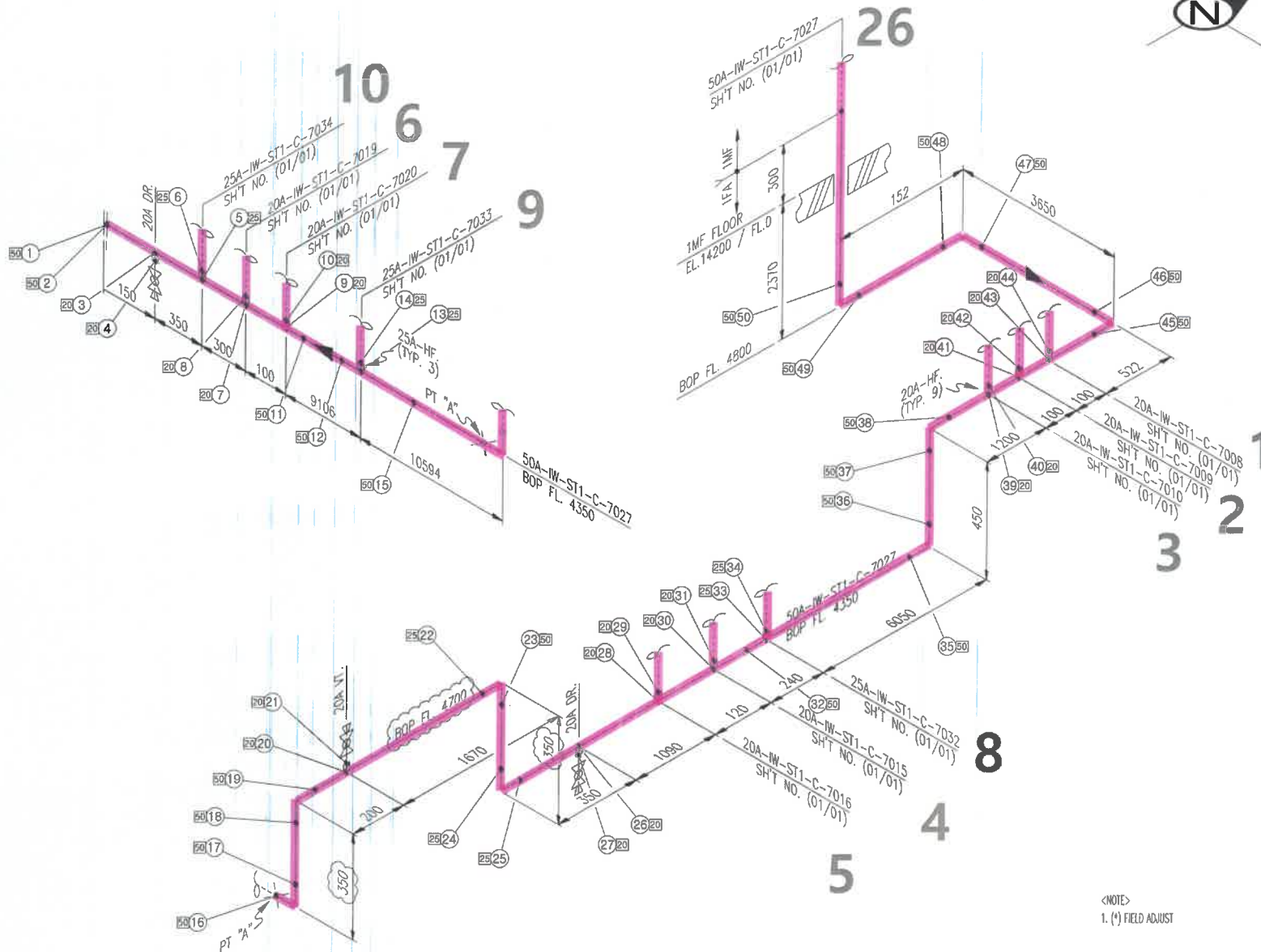
CLIENT

SK hynix

10

CONTRACTOR	SK ecoplant	PROJECT NAME	M15X Ph4 Project
COMMODITY	IW	JOB NO.	22330D
LINE NO.	25A-IW-ST1-C-7034	SH'T NO.	01 / 01
		REV.	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음.



SK ecoplant

ITEM	SIZE	O'TY	DESCRIPTION
ST1			
P	50A	39.8M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	50A	9EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
NP1	20A	3EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
HF	25A	3EA	COUPLING HALF, STS304, SW - SCH 10S
HF	20A	9EA	COUPLING HALF, STS304, SW - SCH 10S
PLG	20A	3EA	PLUG ROUND HEAD, STS304, SCR'D
VB1	20A	3EA	BALL VALVE, KS 10K, SCR'D, BODY-STIS304 / TRIM-STIS304 / SEAT-PTFE, FB, LEVER
F	50A	1EA	FLG SO, KS 10K, RF, STS304
FB	50A	1EA	FLG BLIND, KS 10K, RF, STS304
G	50A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	50A	1SET	MACHINE BOLT & NUTS : STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * SUB-IN : STU * REINFORCING PAD : RP * COMB RED : RC * ECC RED : RE * CON SWAGE : RCS/RCS1 * ECC SWAGE : RCS/RCS1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
 * SUB-IN : STU * SOCKET (HOPE/PMO) : SC * SADDLER HOPE : SD * FLANGE : F/FT * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB10/VB20/VB20L * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VCA/VCS/VCG/VG7
 * GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDO * DAMPER : VP * QUICK COUPLING : SQ * FLEXIBLE JOINT (STEEL) : FLX * FLEXIBLE JOINT (RUBBER) : FLXR * FLEXIBLE JOINT (STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
 * STRAINER (Y-TYPE) : YSTR * STRAINER (T-TYPE) : TSTR * STRAINER (BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : B/B1

INSULATION	CODE	C	CONDITION	PRESS (kg/cm2g)	TEMP (°C)	AREA	W14	UNIT	1FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-B	REF. P & ID	CJ-M15W13-PR-A20-000081	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	10	60										25.01.24	APPROVED FOR CONSTRUCTION	황재중	안남우	박동영
			OPERATING	7.5	AMB														

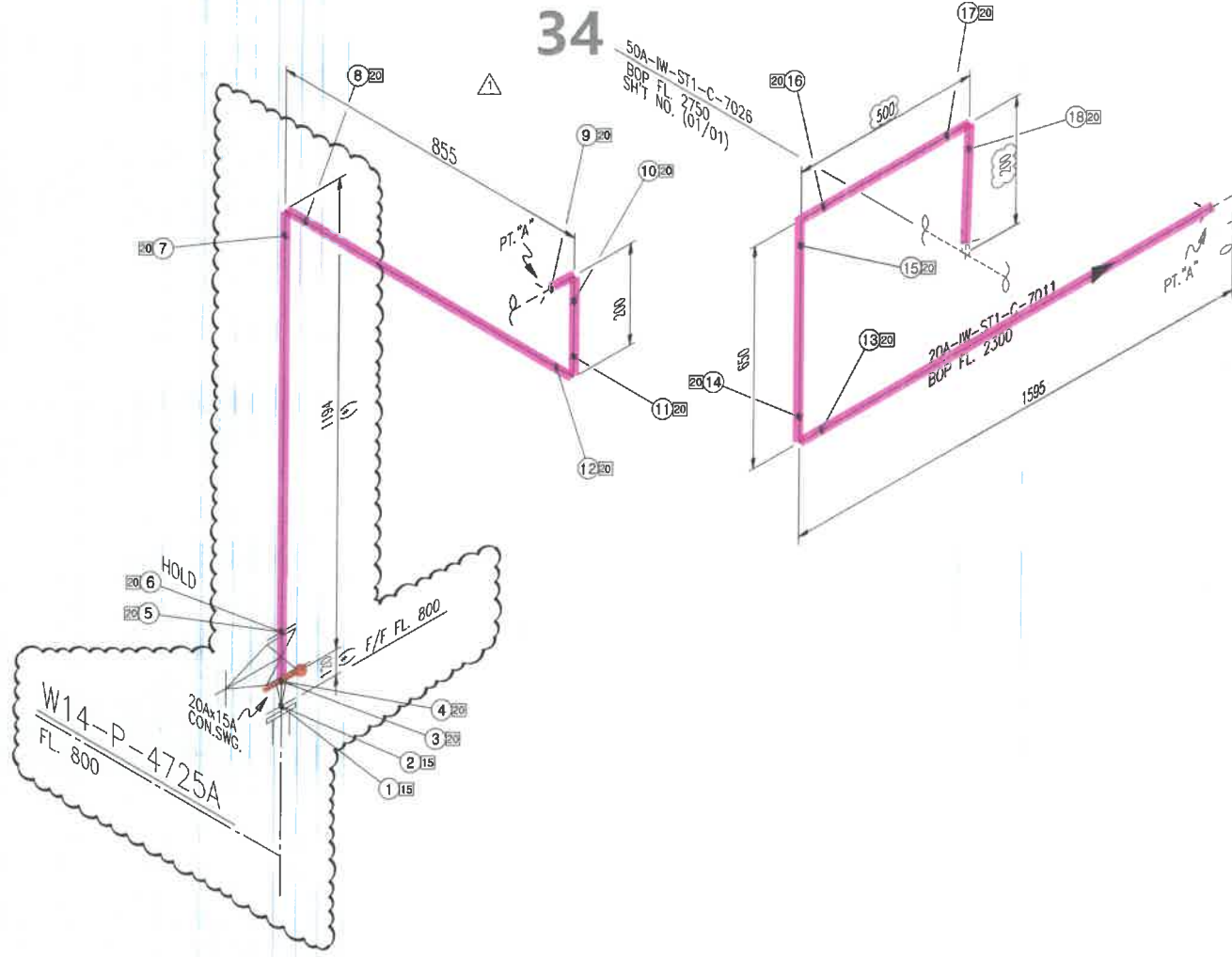
CLIENT

SK hynix

11

CONTRACTOR	SK ecoplant	PROJECT NAME	M15X Ph4 Project
COMMODITY	IW	JOB NO.	22330D
LINE NO.	50A-IW-ST1-C-7027	SH'T NO.	01 / 01
		REV.	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

CLIENT



12

CONTRACTOR



PROJECT NAME

COMMODITY

IW

M15K Ph4 Project

LINE NO.

20A-IW-ST1-C-7011

JOB NO.

22330D

SH'T NO.

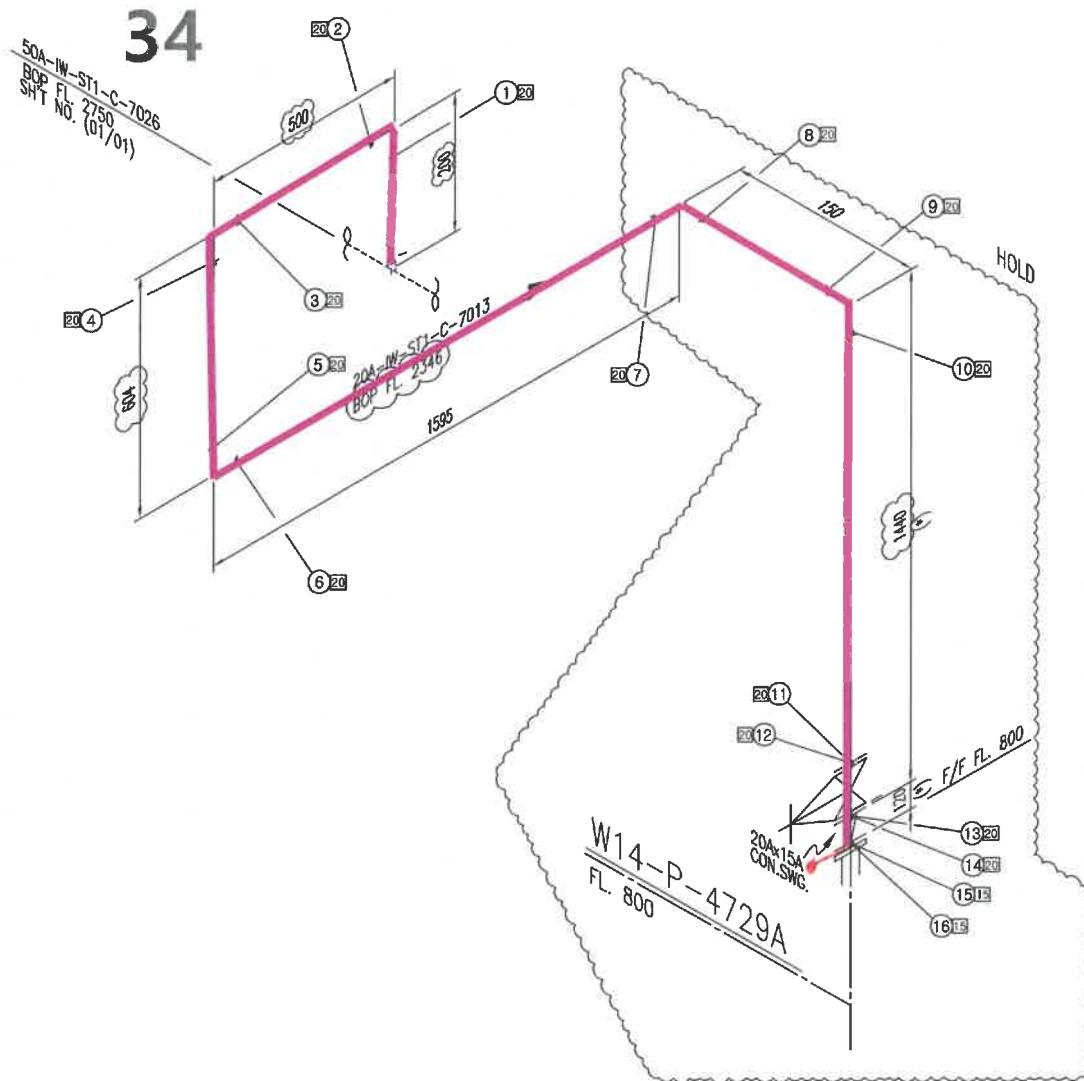
01 / 01

REV.

1

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * REDUCER : TR * STUB-IN : STU * REINFORCING PAD : RP * CONFLUENCE : RC * ECCENTRIC : RE * CONSWAGE : RCS/RCS1 * ECCENTRIC : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET (HDPE/PPVC) : SC * SADDLE (HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VALVE : VG/VG1/VG2 * BALL VALVE : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VALVE : VF/VF1 * CHECK VALVE : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VALVE : VL/VL1/VL2 * DIAPHRAGM VALVE : VD/VD1 * DAMPER : DP * QUICK COUPLING : QC * FLEXIBLE JOINT (STEEL) : FJ * FLEXIBLE JOINT (RUBBER) : FJR * FLEXIBLE JOINT (STEEL PIPE LINE) : FLX1 * FLEXIBLE HOSE : FH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER (Y-TYPE) : YSTR * STRAINER (T-TYPE) : TSTR * STRAINER (BUCKET-TYPE) : BSTR * ANCHOR PLATE : AP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

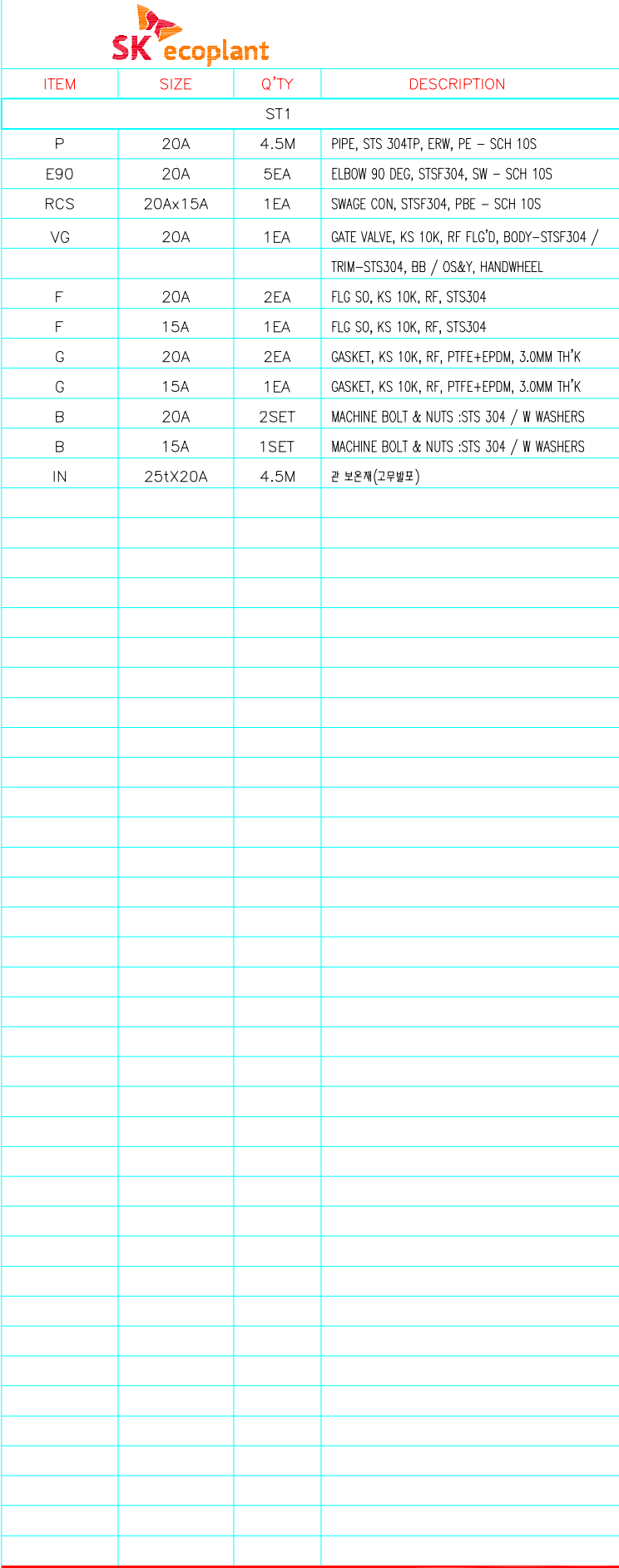
INSULATION	CODE	C	CONDITION	THICK.	DESIGN	OPERATING	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R1MF	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-D	REF. P & ID	CJ-M15W13-PR-A20-000081	REV	DATE	DESCRIPTION	BY	CHK	APP'D
		25					13.5	60														
							10	AMB														



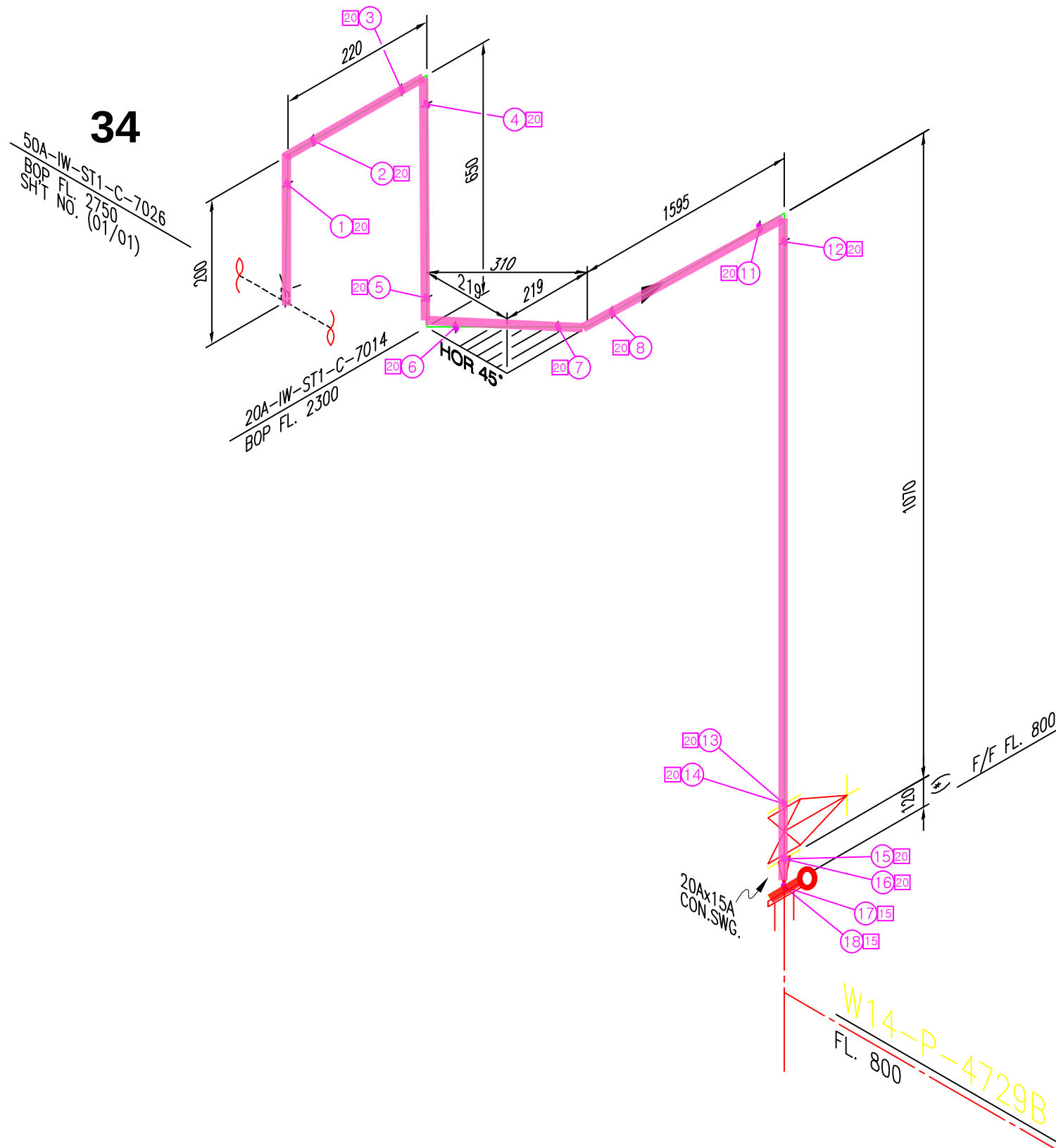
* PFE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* COLLARED : RC	* FOC.RED : RF	* CONSUMAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(DH)PDS : SG	* SOCKET(H)PS : SU	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB2/VB3/VB4/VB5/VB6/VB7/VB8/VB9/VB10/VB11/VB12/VB13/VB14/VB15/VB16/VB17/VB18/VB19/VB20/VB21/VB22/VB23/VB24/VB25/VB26/VB27/VB28/VB29/VB30/VB31/VB32/VB33/VB34/VB35/VB36/VB37/VB38/VB39/VB40/VB41/VB42/VB43/VB44/VB45/VB46/VB47/VB48/VB49/VB50/VB51/VB52/VB53/VB54/VB55/VB56/VB57/VB58/VB59/VB60/VB61/VB62/VB63/VB64/VB65/VB66/VB67/VB68/VB69/VB70/VB71/VB72/VB73/VB74/VB75/VB76/VB77/VB78/VB79/VB80/VB81/VB82/VB83/VB84/VB85/VB86/VB87/VB88/VB89/VB90/VB91/VB92/VB93/VB94/VB95/VB96/VB97/VB98/VB99/VB100/VB101/VB102/VB103/VB104/VB105/VB106/VB107/VB108/VB109/VB110/VB111/VB112/VB113/VB114/VB115/VB116/VB117/VB118/VB119/VB120/VB121/VB122/VB123/VB124/VB125/VB126/VB127/VB128/VB129/VB130/VB131/VB132/VB133/VB134/VB135/VB136/VB137/VB138/VB139/VB140/VB141/VB142/VB143/VB144/VB145/VB146/VB147/VB148/VB149/VB150/VB151/VB152/VB153/VB154/VB155/VB156/VB157/VB158/VB159/VB160/VB161/VB162/VB163/VB164/VB165/VB166/VB167/VB168/VB169/VB170/VB171/VB172/VB173/VB174/VB175/VB176/VB177/VB178/VB179/VB180/VB181/VB182/VB183/VB184/VB185/VB186/VB187/VB188/VB189/VB190/VB191/VB192/VB193/VB194/VB195/VB196/VB197/VB198/VB199/VB200/VB201/VB202/VB203/VB204/VB205/VB206/VB207/VB208/VB209/VB210/VB211/VB212/VB213/VB214/VB215/VB216/VB217/VB218/VB219/VB220/VB221/VB222/VB223/VB224/VB225/VB226/VB227/VB228/VB229/VB230/VB231/VB232/VB233/VB234/VB235/VB236/VB237/VB238/VB239/VB240/VB241/VB242/VB243/VB244/VB245/VB246/VB247/VB248/VB249/VB250/VB251/VB252/VB253/VB254/VB255/VB256/VB257/VB258/VB259/VB260/VB261/VB262/VB263/VB264/VB265/VB266/VB267/VB268/VB269/VB270/VB271/VB272/VB273/VB274/VB275/VB276/VB277/VB278/VB279/VB280/VB281/VB282/VB283/VB284/VB285/VB286/VB287/VB288/VB289/VB290/VB291/VB292/VB293/VB294/VB295/VB296/VB297/VB298/VB299/VB300/VB301/VB302/VB303/VB304/VB305/VB306/VB307/VB308/VB309/VB310/VB311/VB312/VB313/VB314/VB315/VB316/VB317/VB318/VB319/VB320/VB321/VB322/VB323/VB324/VB325/VB326/VB327/VB328/VB329/VB330/VB331/VB332/VB333/VB334/VB335/VB336/VB337/VB338/VB339/VB340/VB341/VB342/VB343/VB344/VB345/VB346/VB347/VB348/VB349/VB350/VB351/VB352/VB353/VB354/VB355/VB356/VB357/VB358/VB359/VB360/VB361/VB362/VB363/VB364/VB365/VB366/VB367/VB368/VB369/VB370/VB371/VB372/VB373/VB374/VB375/VB376/VB377/VB378/VB379/VB380/VB381/VB382/VB383/VB384/VB385/VB386/VB387/VB388/VB389/VB390/VB391/VB392/VB393/VB394/VB395/VB396/VB397/VB398/VB399/VB400/VB401/VB402/VB403/VB404/VB405/VB406/VB407/VB408/VB409/VB410/VB411/VB412/VB413/VB414/VB415/VB416/VB417/VB418/VB419/VB420/VB421/VB422/VB423/VB424/VB425/VB426/VB427/VB428/VB429/VB430/VB431/VB432/VB433/VB434/VB435/VB436/VB437/VB438/VB439/VB440/VB441/VB442/VB443/VB444/VB445/VB446/VB447/VB448/VB449/VB450/VB451/VB452/VB453/VB454/VB455/VB456/VB457/VB458/VB459/VB460/VB461/VB462/VB463/VB464/VB465/VB466/VB467/VB468/VB469/VB470/VB471/VB472/VB473/VB474/VB475/VB476/VB477/VB478/VB479/VB480/VB481/VB482/VB483/VB484/VB485/VB486/VB487/VB488/VB489/VB490/VB491/VB492/VB493/VB494/VB495/VB496/VB497/VB498/VB499/VB500/VB501/VB502/VB503/VB504/VB505/VB506/VB507/VB508/VB509/VB510/VB511/VB512/VB513/VB514/VB515/VB516/VB517/VB518/VB519/VB520/VB521/VB522/VB523/VB524/VB525/VB526/VB527/VB528/VB529/VB530/VB531/VB532/VB533/VB534/VB535/VB536/VB537/VB538/VB539/VB540/VB541/VB542/VB543/VB544/VB545/VB546/VB547/VB548/VB549/VB550/VB551/VB552/VB553/VB554/VB555/VB556/VB557/VB558/VB559/VB560/VB561/VB562/VB563/VB564/VB565/VB566/VB567/VB568/VB569/VB570/VB571/VB572/VB573/VB574/VB575/VB576/VB577/VB578/VB579/VB580/VB581/VB582/VB583/VB584/VB585/VB586/VB587/VB588/VB589/VB590/VB591/VB592/VB593/VB594/VB595/VB596/VB597/VB598/VB599/VB600/VB601/VB602/VB603/VB604/VB605/VB606/VB607/VB608/VB609/VB610/VB611/VB612/VB613/VB614/VB615/VB616/VB617/VB618/VB619/VB620/VB621/VB622/VB623/VB624/VB625/VB626/VB627/VB628/VB629/VB630/VB631/VB632/VB633/VB634/VB635/VB636/VB637/VB638/VB639/VB640/VB641/VB642/VB643/VB644/VB645/VB646/VB647/VB648/VB649/VB650/VB651/VB652/VB653/VB654/VB655/VB656/VB657/VB658/VB659/VB660/VB661/VB662/VB663/VB664/VB665/VB666/VB667/VB668/VB669/VB670/VB671/VB672/VB673/VB674/VB675/VB676/VB677/VB678/VB679/VB680/VB681/VB682/VB683/VB684/VB685/VB686/VB687/VB688/VB689/VB690/VB691/VB692/VB693/VB694/VB695/VB696/VB697/VB698/VB699/VB700/VB701/VB702/VB703/VB704/VB705/VB706/VB707/VB708/VB709/VB710/VB711/VB712/VB713/VB714/VB715/VB716/VB717/VB718/VB719/VB720/VB721/VB722/VB723/VB724/VB725/VB726/VB727/VB728/VB729/VB730/VB731/VB732/VB733/VB734/VB735/VB736/VB737/VB738/VB739/VB740/VB741/VB742/VB743/VB744/VB745/VB746/VB747/VB748/VB749/VB750/VB751/VB752/VB753/VB754/VB755/VB756/VB757/VB758/VB759/VB760/VB761/VB762/VB763/VB764/VB765/VB766/VB767/VB768/VB769/VB770/VB771/VB772/VB773/VB774/VB775/VB776/V							

[illegible][illegible]

CONTRACTOR 		PROJECT NAME	
COMMODITY IW		M15X Pmt Project	
LINE NO. 20A-IW-ST1-C-7013		JOB NO. 22330D	
		SH'T NO. 01	REV. 0



CLIENT		14	
			
CONTRACTOR		PROJECT NAME	
			
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	20A-IW-ST1-C-7013	SH'T NO.	REV.
		01 / 01	0



1. (*) FIELD ADJUST

CLIENT  15

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	13.5	60	UNIT R1MF								M15X Ph4 Project
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-010018-C							COMMODITY IW	JOB NO.
						REF. P & ID CJ-M15W13-PR-A20-000081		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	22330D	
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
												20A-IW-ST1-C-7014	01 / 01	0

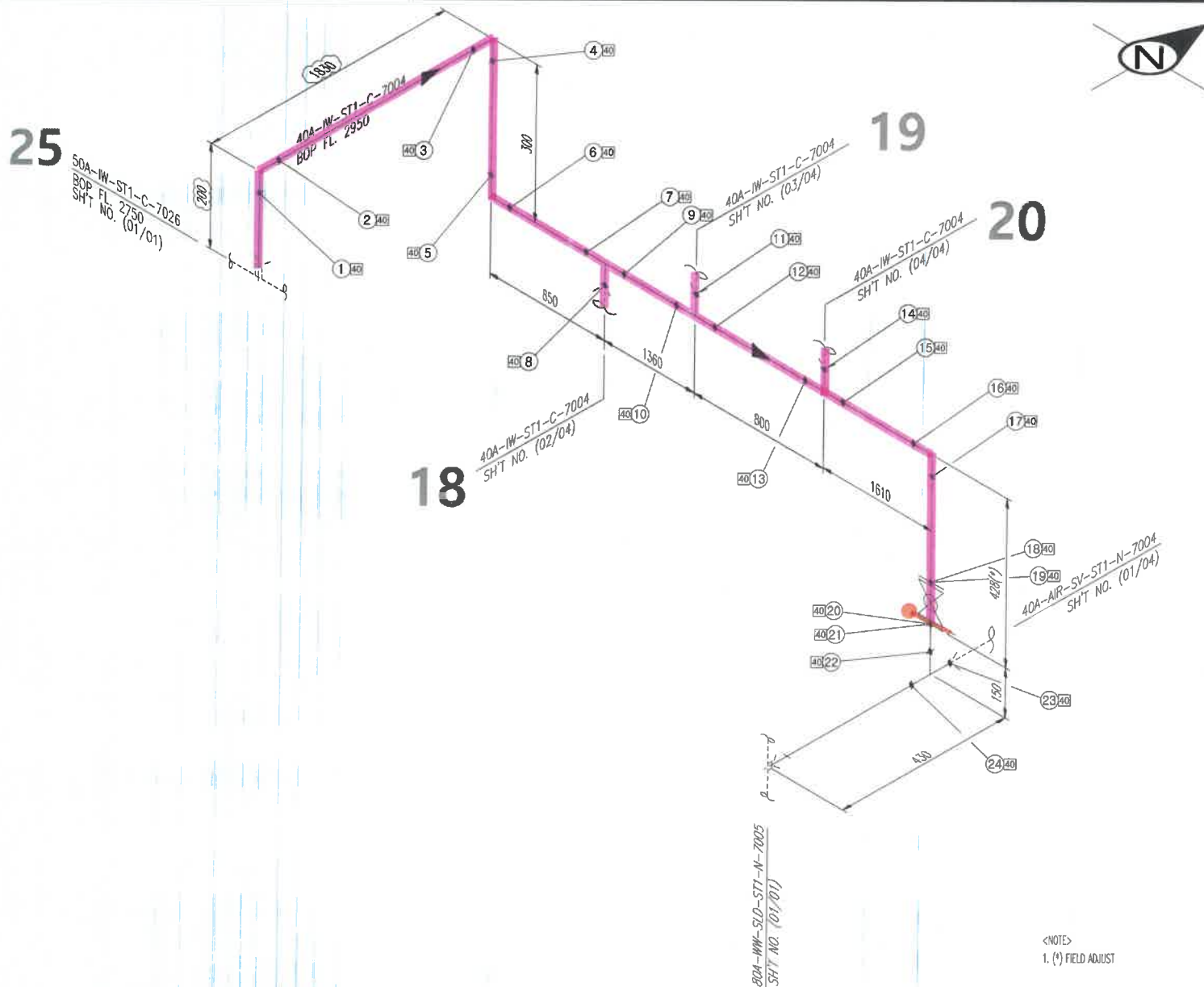


* PMP: P/P1	* 90° ELBOW: E90	* 45° ELBOW: E45	* TEE: T	* RED TEE: TR	* SUB-IN: STU	* REINFORCING PAD: RP	* COULRED: RC	* ECC RED: RE	* CONSWAGE: RCS/RCS1	* ECC SWAGE: RES/RES1	* NIPPLE: NP/NP1/NP2	* CAP: CAP/CAP1	* FLUG: PLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* SUB-IN: STU	* SOCKET(HDPE/HDPE): SC	* SHOULDER(HDPE): SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* RING FLANGE: FB/FB1	* GATE VAL: VG/NG1/NG2	* BALL VAL: VB/VB1/VB2/VB3/VB4	* BUTTERFLY VAL: VF/VF4	* CHECK VAL: VC/NC1/NC2/NC3/NC4/NC5/NC6/NC7					
* GLOBE VAL: VV/VV1/VV2	* DIAPHRAGM VAL: VD/VD1	* DAMPER: VP	* QUICK COUPLING: SQ	* FLEXIBLE JOINT(STEEL): FLJB	* FLEXIBLE JOINT(RUBBER): FLJR	* FLEXIBLE JOINT(STEEL) PTFE LINED: FLJL	* FLEXIBLE HOSE: FLH	* STEAM TRAP: ST	* SPRAY NOZZLE: SPN	* EMERGENCY SHOWER: SSH					
* STRAINER(TYPE): YSTR	* STRAINER(TYPE): TSTR	* STRAINER(BUCKET-TYPE): BSTR	* ANCHOR PLATE: STOP	* BIRD SCREEN: BS	* ISOLATION KIT: IK	* UNLOADING BOX: UB	* AIR TRAP: AT	* LOTO: LOTO	* SIGHT GLASS: SG	* PRESSURE GAUGE: PG	* SHIELD: SH	* SPECIAL BOLT & NUT: RB/RB1			

[illegible]

CLIENT			16		
					
CONTRACTOR			PROJECT NAME		
			M15X Ph4 Project		
COMMODITY			JOB NO.		
IW			223300		
LINE NO.			SH'T NO.		REV.
25A-IW-ST1-C-7028			01 / 01		0

참고: 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 제3자의 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
			ST1
P	40A	B.1M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	4EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	1EA	BALL VALVE, KS 10K, RF FLGD, BODY-STSF304 / TRIM-STSF304 / SEAT-PTFE, FB, LEVER
F	40A	2EA	FLG SO, KS 10K, RF, STS304
G	40A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	40A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
IN	25X40A	B.1M	관 보온재(그루브형)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : SI * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RCS/RCS2 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
 * STUB-IN : SI * SOCKET(HOPE/PMC) : SC * SADDLE(HOPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
 * GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQ * FLEXIBLE JOINT(STEEL) : FLX * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
 * STRAINER(I-TYPE) : YSTR * STRAINER(L-TYPE) : LSTR * STRAINER(BUCKET-TYPE) : BSIR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R1MF	REV	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	25	DESIGN	13.5	60					1	25.03.31	REVISED AS MARKED	황래훈	안남후	박동영
			OPERATING	10	AMB					0	25.01.24	APPROVED FOR CONSTRUCTION	황래훈	안남후	박동영
												REF. PLAN DWG.			
												CJ-M15W14-PI-A30-010018-C			
												REF. P & ID			
												CJ-M15W14-PR-A20-000502			

CLIENT

SK hynix

17

CONTRACTOR

SK ecoplant

PROJECT NAME

M15X Ph4 Project

COMMODITY

IW

JOB NO.

22330D

LINE NO.

40A-IW-ST1-C-7004

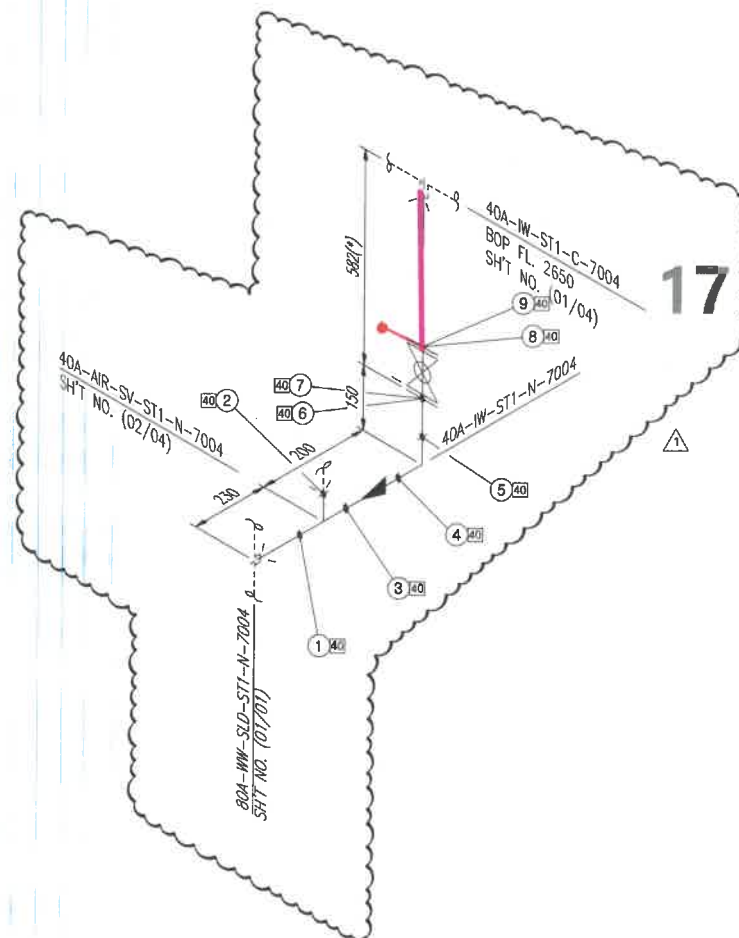
SH'T NO.

01 / 04

REV.

1

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (4) FIELD ADJUST

* PIPE: P/P1	* 90° ELBOW: F90	* 45° ELBOW: F45	* TEE: T	* RED TEE: TR	* STUB-IN: STU	* REINFORCING PAD: RP	* CONJRED: RC	* ECC RED: RE	* CON SWAGE: RES/WS1	* ECC SWAGE: RES/WS1	* NIPPLE: NP/NP1/NP2	* CAP: CAP/CAP1	* PLUG: PLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* STUB-IN: SI	* SOCKET/HPIPE/PWPE: SC	* SADDLE/HPIPE: SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* RIND FLANGE: FB/FB1	* GATE VA: VG/VG1/VG2	* BALL VA: VB/VB1/VB2/VB3/VB4	* BUTTERFLY VA: VF/VF1	* CHECK VA: VC/VCA/VCB/VCC/VCA1/VCA2/VCA3/VCA4/VCA5/VCA6/VCA7	* SPRAY NOZZLE: SN	* EMERGENCY SHOWER: SSH			
* GLOBE VA: VL/VL1/VL2	* DIAPHRAGM VA: VD/VD1	* DAMPER: VP	* QUICK COUPLER: Q	* FLEXIBLE JOINT(STEEL): FJ	* FLEXIBLE JOINT(RUBBER): FLJR	* FLEXIBLE JOINT(STEEL) PTFE LINED: FLXLP	* FLEXIBLE HOSE: FH	* STEAM TRAP: ST	* SPRAY NOZZLE: SN						
* STRAINER(N-TYPE): YSTR	* STRAINER(T-TYPE): TSTR	* STRAINER(BLOCK-TYPE): BSTR	* ANCHOR PLATE: SLP	* BIRD SCREEN: BS	* ISOLATION TUB: IB	* UNLOADING BOSS: UB	* AIR TRAP: AT	* LOTO: LOTO	* SIGHT GLASS: SG	* PRESSURE GAUGE: PG	* SHIELD: SH	* SPECIAL BOLT: NUT: B/NUT1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	13.5	60	UNIT R1MF								M15X Ph4 Project
			OPERATING	10	AMB	REF. PLAN DWG.		25.03.31	REVISED AS MARKED	황재동	안남후	박종영	COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-010018-C		25.01.24	APPROVED FOR CONSTRUCTION	황재동	안남후	박종영	IW	22330D
						REF. P & ID	REV.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.
						CJ-M15W14-PR-A20-000502							40A-IW-ST1-C-7004	02 / 04
														REV. 1

경고: 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



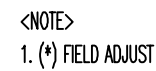
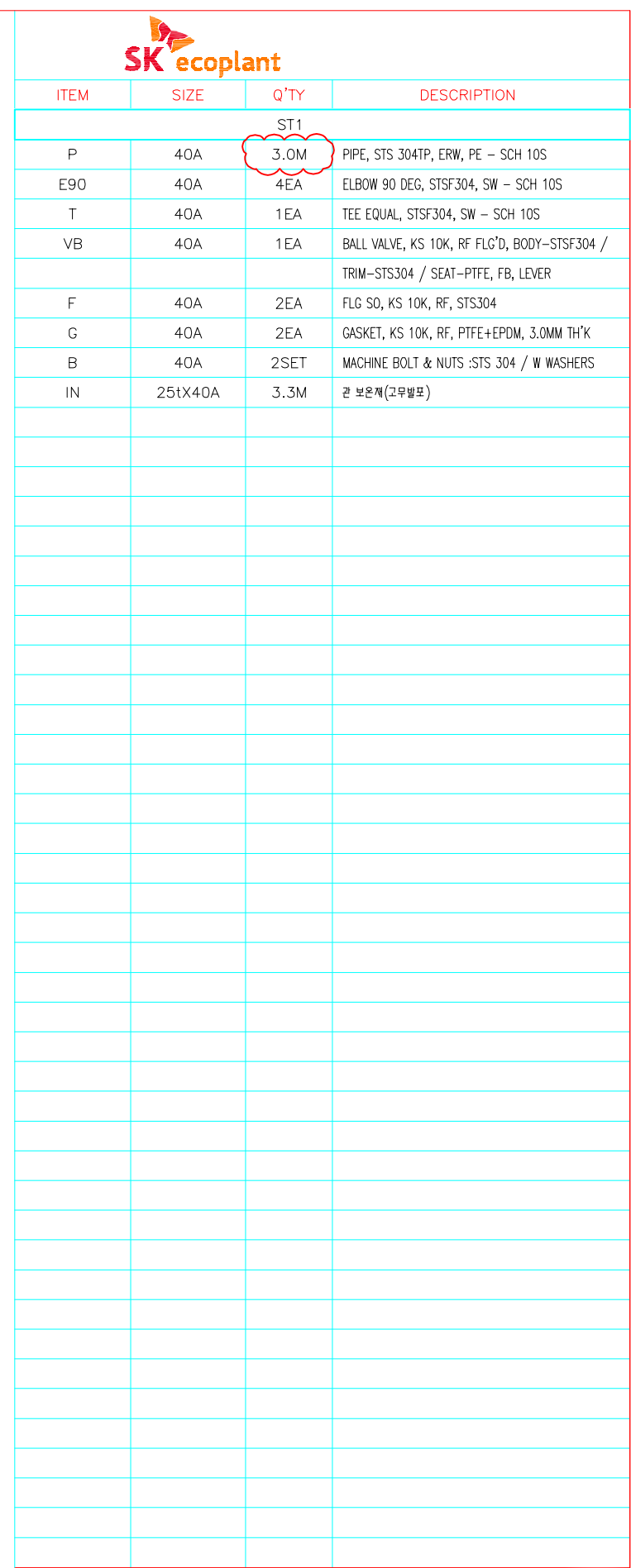
CLIENT



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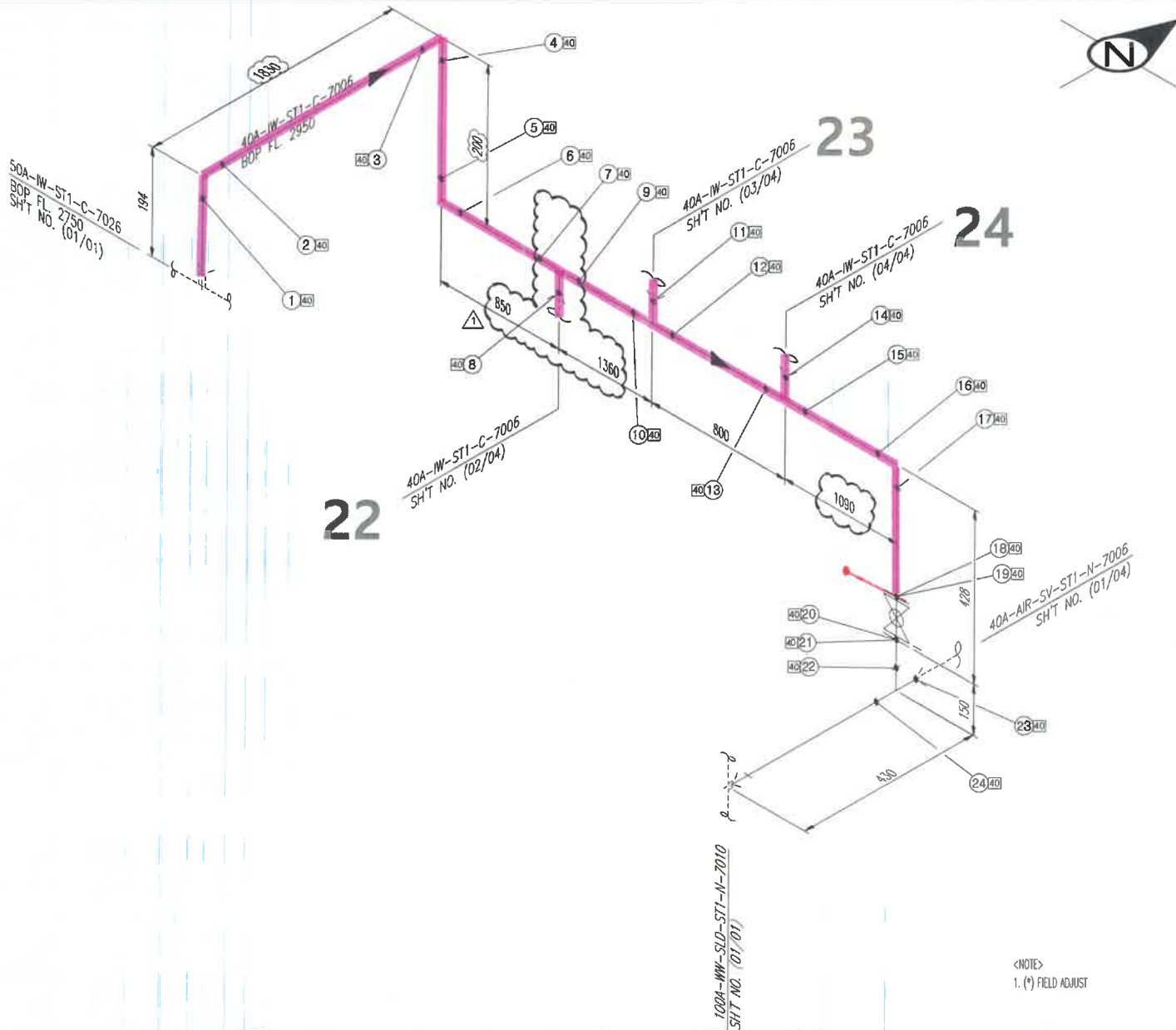
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	40A-IW-ST1-C-7004	SH'T NO.	REV.
		03 / 04	0

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	25	DESIGN	13.5	60	UNIT	R1MF						
			OPERATING	10	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-010018-C							
						REF. P. & ID							
						CJ-M15W14-PR-A20-000502							
									25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'



CLIENT  20

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	13.5	60	UNIT	R1MF							M15X Ph4 Project		
			OPERATING	10	AMB	REF. PLAN DWG.							COMMODITY	IW	JOB NO.	
						CJ-M15W14-PI-A30-010018-C			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	40A-IW-ST1-C-7004	SH'T NO.
						CJ-M15W14-PR-A20-000502	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			04 / 04	REV.



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON. RED : RC * ECC. RED : RE * CON. SWAGE : RCS/RCS1 * ECC. SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PMC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1 * DAMPER : VP * QUICK COUPLING : SOC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSIR * ANCHOR PLATE : STOP * BRD. SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOT0 : LOT0 * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14	UNIT	R1MF	REF. PLAN DWG.	CJ-M15W14-PI-A30-010018-C	REF. P & ID	CJ-M15W14-PR-A20-000503	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	25	DESIGN	13.5	60									25.03.31	REVISED AS MARKED	황래용	안남후	박동영
			OPERATING	10	AMB									25.01.24	APPROVED FOR CONSTRUCTION	황래용	안남후	박동영

SK ecoplant

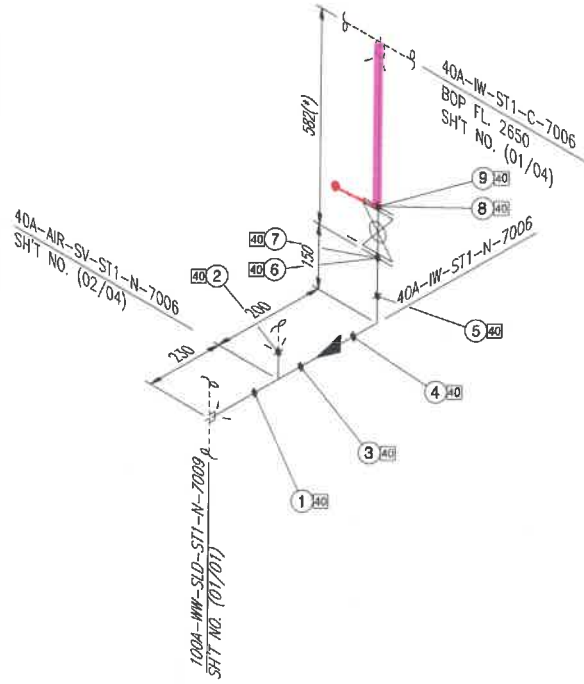
ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	40A	8.1M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	4EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 / TRIM-STSF304 / SEAT-PTFE, FB, LEVER
F	40A	2EA	FLG SO, KS 10K, RF, STSF304
G	40A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	40A	2SET	MACHINE BOLT & NUTS : STS 304 / W WASHERS
IN	25X40A	8.1M	관 보온재(가우덴트)

CLIENT

SK hynix

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CONTRACTOR	SK ecoplant	PROJECT NAME	M15X Ph4 Project
COMMODITY	IW	JOB NO.	22330D
LINE NO.	40A-IW-ST1-C-7006	SH'T NO.	01 / 04
		REV.	1



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<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

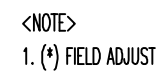
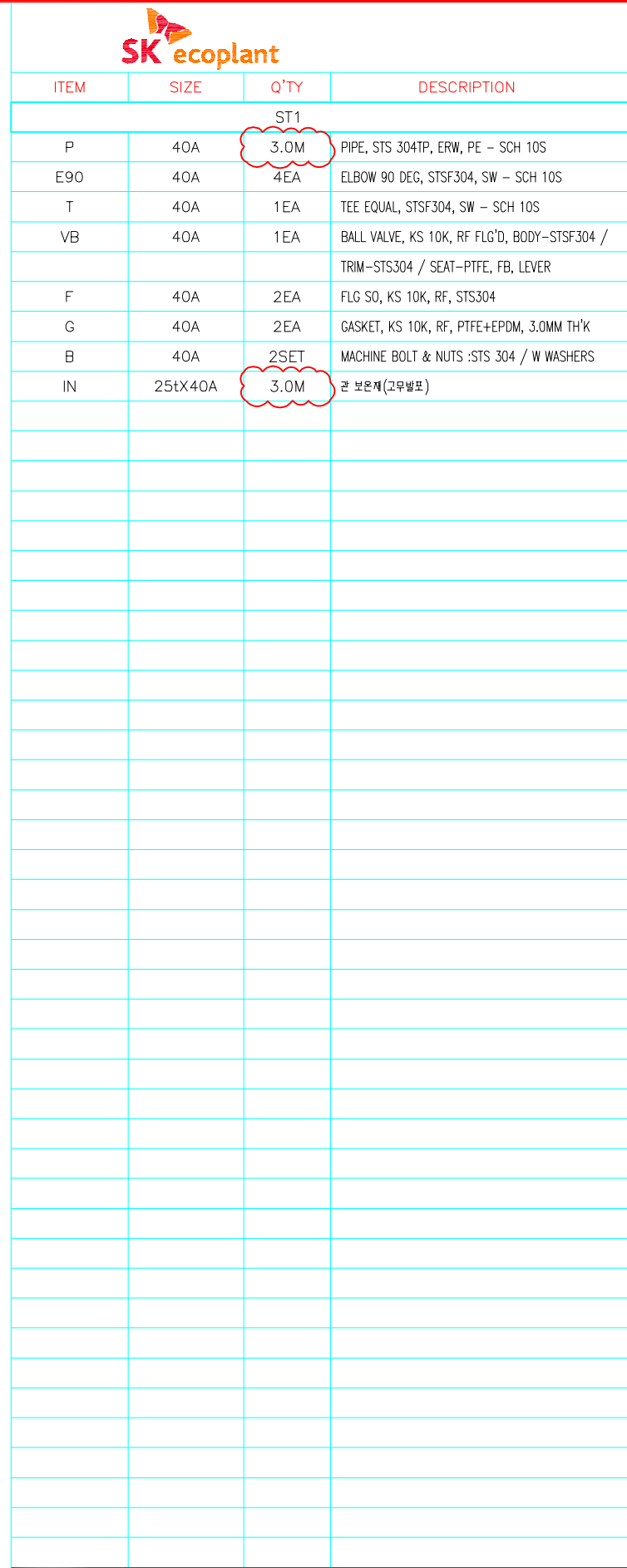
ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	1.1M	PIPE, STS 304TP, ERM, PE - SCH 10S
E90	40A	1EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	1EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	1EA	BALL VALVE, KS 10K, RF FLGD, BODY-STSF304 /
			TRIM-STSF304 / SEAT-PTFE, FB, LEVER
F	40A	2EA	FLG SO, KS 10K, RF, STSF304
G	40A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	40A	2SET	MACHINE BOLT & NUTS -STS 304 / W WASHERS
IN	251X40A	1.1M	관 보온재(그루보온)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RCS/RCS1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PMC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VG1/VG2/VG3/VG4/VG5/VG6/VG7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQ * FLEXIBLE JOINT(STEEL) : FLX * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(NY-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/B1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	RIMF	REF. PLAN DWG.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	25	DESIGN	13.5	60					CJ-M15W14-PI-A30-010018-C	25.03.31	REVISED AS MARKED	황래준	한남후	박동영
			OPERATING	10	AMB					REF. P & ID	25.01.24	APPROVED FOR CONSTRUCTION	황래준	한남후	박동영
										CJ-M15W14-PR-A20-000503					

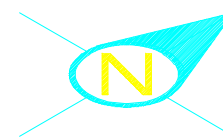
CLIENT	SK hynix	22
CONTRACTOR	SK ecoplant	PROJECT NAME
COMMODITY	IW	M15X Ph4 Project
LINE NO.	40A-IW-ST1-C-7006	JOB NO.
		223300
		SHT NO. REV.
		02 / 04 1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음.



CLIENT  23

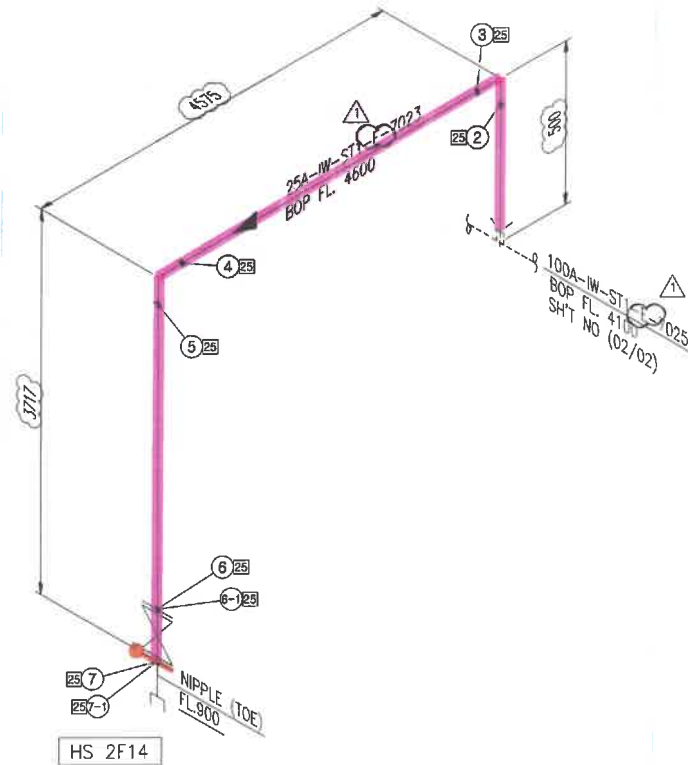
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CLIENT  25

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14						
	THICK.	-	DESIGN	10	60	UNIT 1MF						
			OPERATING	7.5	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-010017-D		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
						REF. P & ID CJ-M15W13-PR-A20-000081		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'

CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY IW		JOB NO. 22330D	
LINE NO. 50A-IW-ST1-C-7026	SH'T NO. 01 / 01		REV. 1



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<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : F90	* 45° ELBOW : F45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* COLLARED : RC	* EGG.RED : RE	* CONSUMAGE : RCS/RCS1	* ECG.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLC	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET/THEFT-PIPE : SC	* SADDLE(THEFT) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VAL : VG/VG1/VG2	* BALL VAL : VB/VB1/VB2/VB3/VB4	* BUTTERFLY VAL : VF/VF1	* CHECK VAL : VC/VG1/VG2/VG3/VG4/VG5/VG6/VG7	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH		
* GLOBE VAL : VL/VL1/VL2	* DIAPHRAGM VAL : VD/VD1	* DAMPER : DP	* QUICK COUPLING : SQ	* FLEXIBLE JOINT(STEEL) : FLXJ	* ISOLATION KIT(RUBBER) : IKR	* FLEXIBLE JOINT(STEEL) : FLXJ	* FLEXIBLE JOINT(FLEX LINECD) : FLXJ	* FLEXIBLE HOSE : FHJ	* STEAM TRAP : ST	* PRESSURE GAUGE : PG	* SHIELD : SH	* SPECIAL BOLT & NUT : BL/B1			
* STRAINER(TYPE) : YSTR	* STRAINER(TYPE) : YSTR	* CHIMNEY(BLOCK-TYPE) : YSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* FLOUNDER KOT : FK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG						

[illegible][illegible]

CLIENT



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CONTRACTOR	
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PROJECT NAME
M15X Ph4 Project

COMMODITY	
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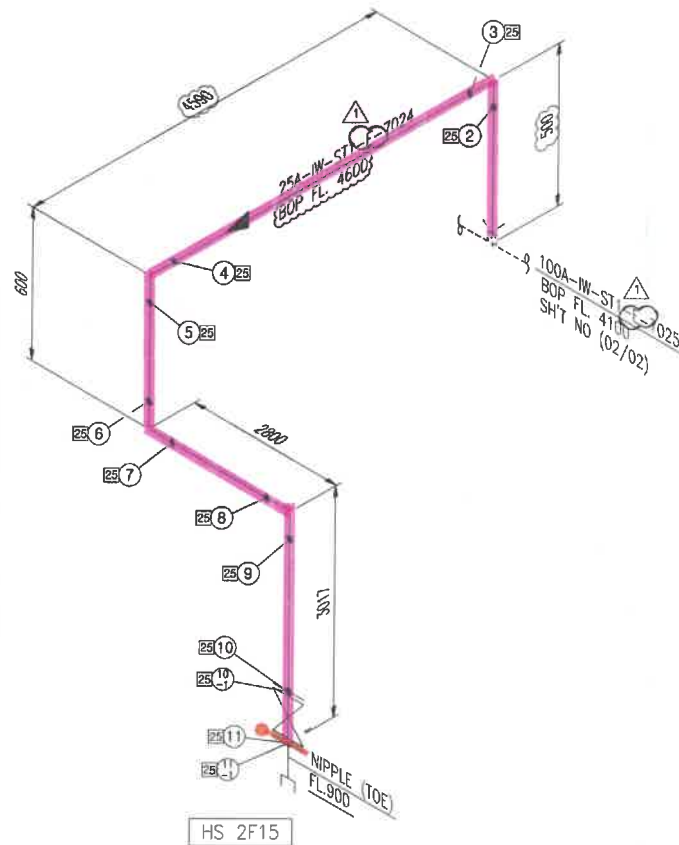
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LINE NO.	
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25A-W-STC-E-7023

JOB NO. 22330D

SH'T NO.	REV.
01 / 01	1



36

<NOTE>
1. (*) FIELD ADJUST

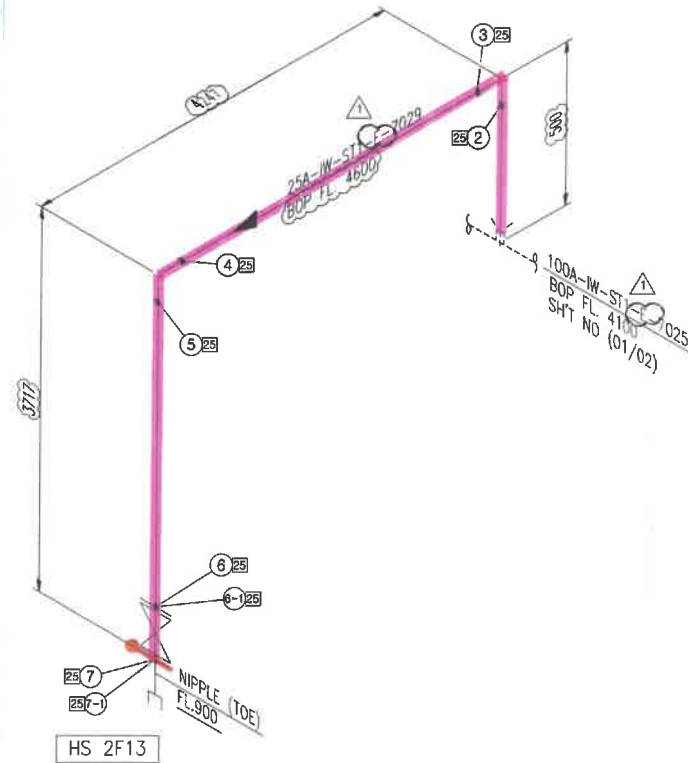
SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	25A	11.0M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	4EA	ELBOW 90 DEG, STS304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STIS304 / TRIM-STIS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(CABLE)
INC	25IX25A	11.0M	관 보강재(코팅필름) + 알루미늄 CASING

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STEEL-IN : ST1 * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STEEL-IN : ST1 * SOCKET(HDPE/PPVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1/VD2 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXG * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STAP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTD : LOTD * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : RL/RL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R2FA	REF. PLAN DWG.	REV.	DATE	DESCRIPTION	BY	CHK	APP'D	CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	13.5	60					REF. PLAN DWG.	1	25.03.31	REVISED AS MARKED	황래용	안남후	박동영	SK ecoplant	M15X Ph4 Project
			OPERATING	10	AMB					CJ-M15W14-PI-A30-020016-B	0	25.01.24	APPROVED FOR CONSTRUCTION	황래용	안남후	박동영		JOB NO.
										REF. P & ID								223300
										CJ-M15W13-PR-A20-000081								SH'T NO.
																		REV.
																		01 / 01
																		1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유무자의 불법 사용으로 피해가 발생할 경우 유무자에게 인/형사상 책임을 요구할 수 있음.



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<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	25A	8.3M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 / TRIM-STSF304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SOC	25A	1EA	SOCKET (HOPE)
1 INC	25IX25A	8.3M	관 브랜치 (관망관) + 일체형 CASING 1

CLIENT

SK hynix

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CONTRACTOR

SK ecoplant

PROJECT NAME

M15X Ph4 Project

COMMODITY

IW

JOB NO.

22330D

LINE NO.

25A-1W-ST-1029

SH'T NO.

01 / 01

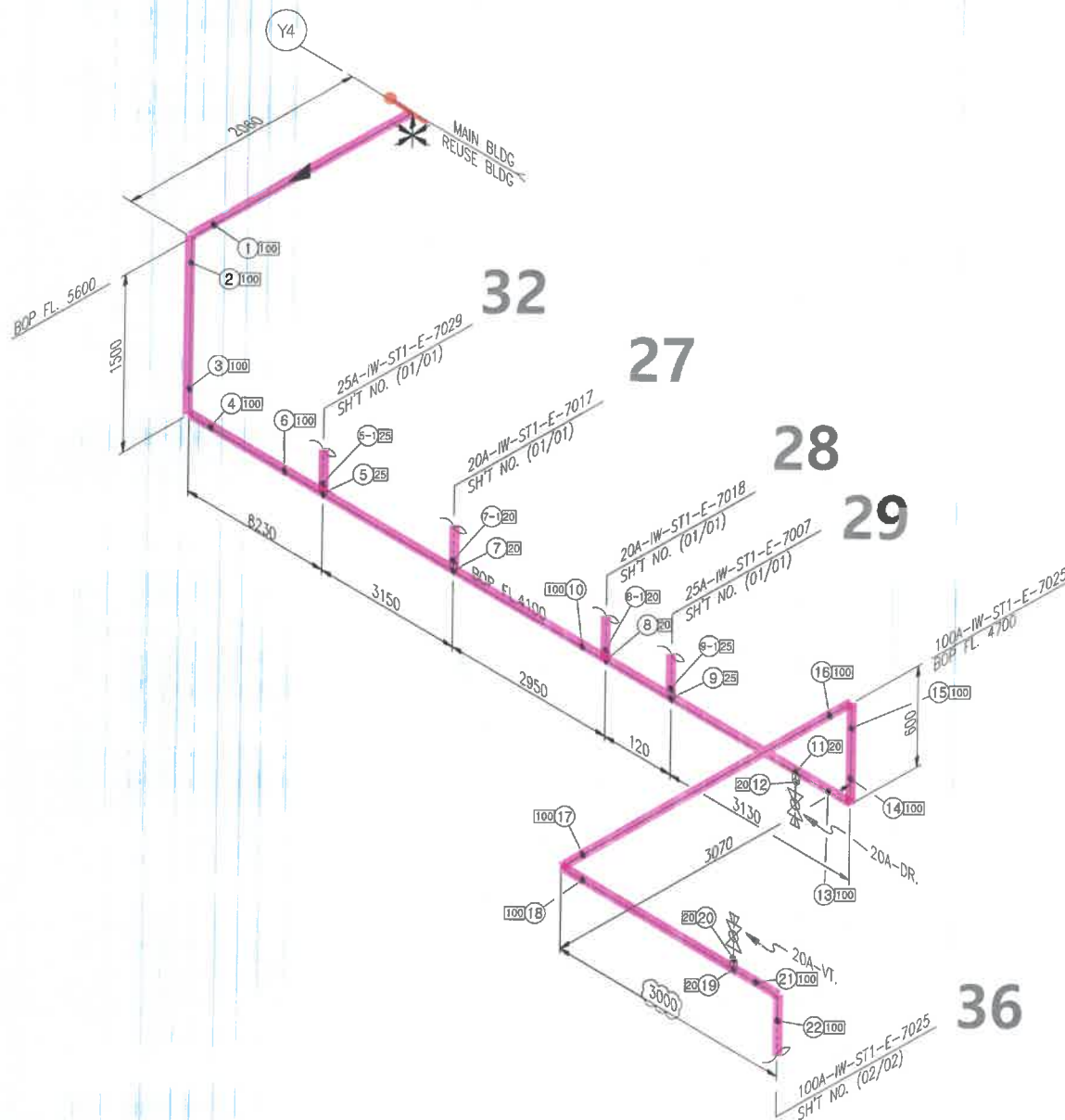
REV.

1

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : SIU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/BES1 * ECC.SWAGE : RCS/BES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/H11
* STUB-IN : SIU * SOCKET (HOPE/PWC) : SC * SADDLE (HOPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DAMPHASH VA : VO/VOLO * DAMPER : VP * QUICK COUPLING : SOC * FLEXIBLE JOINT (STEEL) : FLXB * FLEXIBLE JOINT (RUBBER) : FLXR * FLEXIBLE JOINT (STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER (Y-TYPE) : YSTR * STRAINER (I-TYPE) : ISTR * STRAINER (BUCKET-TYPE) : BSIR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTD : LOTD * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/B11

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14	UNIT	R2FA	REF. PLAN DWG.	REF. P & ID	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	25	DESIGN	13.5	60					CJ-M15W14-PI-A30-020014-B		25.03.31	REVISED AS MARKED	홍래중	안남후	박동영
			OPERATING	10	AMB							25.01.24	APPROVED FOR CONSTRUCTION	홍래중	안남후	박동영

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	100A	27.9M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	100A	6EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
NP1	20A	2EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
HF	25A	2EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	20A	4EA	COUPLING HALF, STSF304, SW - SCH 10S
VB1	20A	2EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 / TRIM-ST304 / SEAT-PTFE, FB, LEVER
PLG	20A	2EA	PLUG ROUND HEAD, STSF304, SCR'D
INC	25X100A	27.9M	콘 보온재(고무발포) + 발수피막 CASING

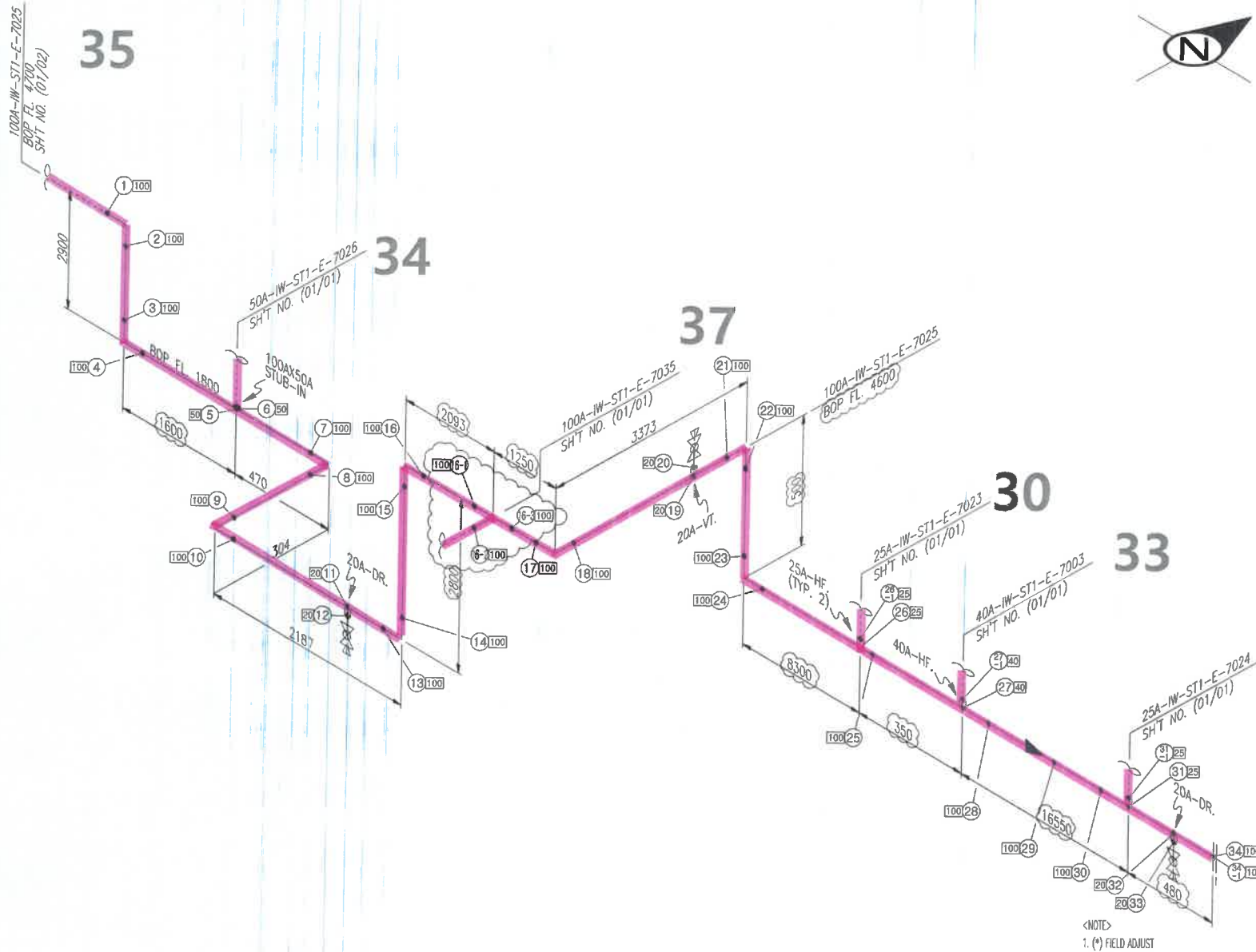
<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
 * STUB-IN : STU * SOCKET(HOPE/PMC) : SC * SADDLE(HOPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA. : VF/VF1 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLX8 * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLX1 * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
 * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/NL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14
	THICK.	25	DESIGN	10	60	UNIT	2FA
			OPERATING	7.5	AMB	REF. PLAN DWG.	
						CJ-M15W14-PI-A30-010017-D	
						REF. P & ID	
						CJ-M15W13-PR-A20-000081	

CLIENT	SK hynix	35
CONTRACTOR	SK ecoplant	PROJECT NAME
COMMODITY	IW	M15X P14 Project
LINE NO.	100A-IW-ST1-E-7025	JOB NO.
		223300D
		SH'T NO.
		01 / 02
		REV.
		0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 복제 및 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음.



SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	100A	43.4M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	100A	9EA	ELBOW 90 DEG UR, STS304, ERW, BE - SCH 10S
T	100A	1EA	TEE EQUAL, STS304, SW - SCH 10S
NP1	20A	3EA	NIPPLE, STS 304TP, ERW, TDE - SCH 40S
STU	100AX50A	1EA	STUB-IN BRANCH
HF	40A	1EA	COUPLING HALF, STS304, SW - SCH 10S
HF	25A	2EA	COUPLING HALF, STS304, SW - SCH 10S
HF	20A	3EA	COUPLING HALF, STS304, SW - SCH 10S
VB1	20A	3EA	BALL VALVE, KS 10K, SCR'D, BODY-STIS304 / TRIM-STIS304 / SEAT-PTFE, FB, LEVER
PLG	20A	3EA	PLUG ROUND HEAD, STS304, SCR'D
F	100A	1EA	FLG SO, KS 10K, RF, STS304
FB	100A	1EA	FLG BLIND, KS 10K, RF, STS304
G	100A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	100A	1SET	MACHINE BOLT & NUTS-STIS 304 / W WASHERS
INC	25X100A	43.4M	본 보른체(구멍별) + 일체형 CASING

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * DOWNLED : RC * ECC RED : RE * CON SWAGE : RCS/RCS1 * ECC SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
 * STUB-IN : STU * SOCKET(HDPE/PPVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
 * GLOBE VA : VV/VV1/VV2 * DIAPHRAGM VA : VD/VD1/VD2 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
 * STRAINERY(TYPE) : YSTR * STRAINER(TYPE) : YSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IS * UNLOADING BOX : UB * AIR TRAP : AT * LOT : LOT1 * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/HL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm ²)	TEMP. (°C)	AREA	W14
	THICK.	25	DESIGN	10	60	UNIT	2FA
			OPERATING	7.5	AMB	REF. PLAN DWG.	
						CJ-M15W14-PI-A30-020015-A	
						REF. P & ID	
						CJ-M15W13-PR-A20-000081	

CLIENT

SK hynix

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CONTRACTOR

SK ecoplant

COMMODITY

IW

LINE NO.

100A-IW-ST1-E-7025

PROJECT NAME

M15K Ph4 Project

JOB NO.

22330D

SHT NO.

02 / 02

REV.

0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유출자의 불법 사용으로 인하여 발생할 경우 유출자에게 민/형사상 책임을 요구할 수 있음.



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CLIENT		 37	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	100A-IW-S11-E-7035	SH'T NO. 01 / 01	REV. 0

<NOTE>
1. (*) FIELD ADJUST

경고: 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

PUNCH LIST

Project No : 22330D

Project Title : M15X Ph3 Project

Package No :

Date :

S/N	Drawing No	Description	Category A B C	Punch Clear Status			
				작성자		확인자	
				Name	Sign	Name	Sign
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Remarks

A : Before Hydro Test

B : Before M/C

C : Before Commissioning